```
FUNCTION BAPI_DELIVERY_NETWORK_CHANGE.
*"----------------------------------------------------------------------
*"*"Local interface:
*"  IMPORTING
*"     VALUE(BAPIOIUDNHEADER) LIKE  BAPI_OIU_DN STRUCTURE  BAPI_OIU_DN
*"     VALUE(BAPIOIUDN1DETAIL) LIKE  BAPI_OIU_DN_1 STRUCTURE
*"        BAPI_OIU_DN_1
*"  EXPORTING
*"     VALUE(BAPIOIUDNHEADERE) LIKE  BAPI_OIU_DN STRUCTURE  BAPI_OIU_DN
*"     VALUE(BAPIOIUDN1DETAILE) LIKE  BAPI_OIU_DN_1 STRUCTURE
*"        BAPI_OIU_DN_1
*"  TABLES
*"      RETURN STRUCTURE  BAPIRET2
*"----------------------------------------------------------------------

DATA:  BEGIN OF ORIG_OIU_PR_DN OCCURS 1.
         INCLUDE STRUCTURE ROIU_PR_DN.
DATA:  END   OF ORIG_OIU_PR_DN.

DATA:  CHANGE_OIU_PR_DN LIKE ORIG_OIU_PR_DN.
* Clear message table and set bapi flag
  REFRESH G_RETURN_TB.
* CLEAR ALL WORKING STORAGE.
  PERFORM CLEAR_INFORMATION.

  CLEAR G_BAPI_STOP.
  G_BAPI_FG = C_TRUE.

* Authority check
  perform check_bapi_authorization using 'O3U_DN2'.

  MOVE-CORRESPONDING BAPIOIUDNHEADER  TO CHANGE_OIU_PR_DN.
  MOVE-CORRESPONDING BAPIOIUDN1DETAIL TO CHANGE_OIU_PR_DN.
  CHANGE_OIU_PR_DN-BUKRS = BAPIOIUDN1DETAIL-COMP_CODE.
  CHANGE_OIU_PR_DN-VNAME = BAPIOIUDN1DETAIL-VENTURE.

* changes all field to uppercase except for description
  PERFORM TRANSLATE_TEXTS.
* Get the delivery network detail information
  PERFORM GET_REF_DELIVERY_NETWORK USING CHANGE_OIU_PR_DN-DN_NO
                                         TEXT-002
                                         CHANGE_OIU_PR_DN-DN_NO
                                         SPACE
                                         SPACE.
* If the delivery network does not exist
  CHECK_BAPI_STOP.
* Move the old values from the structure into temporary internal table
  MOVE-CORRESPONDING ROIU_PR_DN TO ORIG_OIU_PR_DN.
* Move the new values into the old structure for edits and validations
  CHANGE_OIU_PR_DN-DN_NO   = ROIU_PR_DN-DN_NO.
  CHANGE_OIU_PR_DN-TYPE_CD = ROIU_PR_DN-TYPE_CD.
  CHANGE_OIU_PR_DN-DN_GRP  = ROIU_PR_DN-DN_GRP.
  MOVE-CORRESPONDING CHANGE_OIU_PR_DN TO ROIU_PR_DN.

  IF ORIG_OIU_PR_DN NE CHANGE_OIU_PR_DN.
*   validate the fields required
    PERFORM BAPI_EDIT_CHECKS_CHANGE.
*   If edits and validations have an error pass messages to user
    CHECK_BAPI_STOP.
*   Lock the record from changes
    PERFORM LOCK_DELIVERY_NUMBER.
```

```
*   Enqueue of DN Standard Hierarchy                    " Start SOGK013053 EY
    IF ORIG_OIU_PR_DN-dn_set <> CHANGE_OIU_PR_DN-dn_set.
       PERFORM enqueue_setid USING ORIG_OIU_PR_DN-dn_set
                                   CHANGE_OIU_PR_DN-dn_set.
    ENDIF.
    CHECK_BAPI_STOP.                                    " End   SOGK013053 EY
*   Prepare all the fields for the save.
    PERFORM PREPARE_UPDATE_FOR_SAVE.
*   Call the function to update the table
    PERFORM SAVE_DEL_NET_DATA.
    Perform F_UPDATE_HIERARCHY.                            "SOGK027671 VV
    CHECK_BAPI_STOP.
*   Call the function to update the change document
    PERFORM CHANGE_DOCUMENT USING ORIG_OIU_PR_DN
                                  ROIU_PR_DN.
*   Error message of the change document failed.
    CHECK_BAPI_STOP.
*   Sends user a success message
    PERFORM SAVE_SUCCESSFUL.
*   Return success message from the save and exits the function module
    RETURN[] = G_RETURN_TB[].
*   Move the new records to the return structure
    MOVE-CORRESPONDING ROIU_PR_DN TO BAPIOIUDNHEADERE.
    MOVE-CORRESPONDING ROIU_PR_DN TO BAPIOIUDN1DETAILE.
  ENDIF.

ENDFUNCTION.
```

Subroutines:		Line	Page
BAPI_EDIT_CHECKS_CHANGE.....	LOIU_DELIVERY_NETWORKU04	57	1
CHANGE_DOCUMENT.....	LOIU_DELIVERY_NETWORKU04	75	2
CHECK_BAPI_AUTHORIZATION.....	LOIU_DELIVERY_NETWORKU04	30	1
CLEAR_INFORMATION.....	LOIU_DELIVERY_NETWORKU04	24	1
ENQUEUE_SETID.....	LOIU_DELIVERY_NETWORKU04	64	2
F_UPDATE_HIERARCHY.....	LOIU_DELIVERY_NETWORKU04	72	2
GET_REF_DELIVERY_NETWORK.....	LOIU_DELIVERY_NETWORKU04	40	1
LOCK_DELIVERY_NUMBER.....	LOIU_DELIVERY_NETWORKU04	61	1
PREPARE_UPDATE_FOR_SAVE.....	LOIU_DELIVERY_NETWORKU04	69	2
SAVE_DEL_NET_DATA.....	LOIU_DELIVERY_NETWORKU04	71	2
SAVE_SUCCESSFUL.....	LOIU_DELIVERY_NETWORKU04	80	2
TRANSLATE_TEXTS.....	LOIU_DELIVERY_NETWORKU04	38	1

```
FUNCTION BAPI_DELIVERY_NETWORK_CREATE.
*"----------------------------------------------------------------------
*"*"Local interface:
*"       IMPORTING
*"             VALUE(BAPIOIUDNHEADER) LIKE  BAPI_OIU_DN
*"                                    STRUCTURE  BAPI_OIU_DN
*"             VALUE(BAPIOIUDN1DETAIL) LIKE  BAPI_OIU_DN_1
*"                                    STRUCTURE  BAPI_OIU_DN_1
*"       EXPORTING
*"             VALUE(BAPIOIUDNHEADERE) LIKE  BAPI_OIU_DN
*"                                    STRUCTURE  BAPI_OIU_DN
*"             VALUE(BAPIOIUDN1DETAILE) LIKE  BAPI_OIU_DN_1
*"                                    STRUCTURE  BAPI_OIU_DN_1
*"       TABLES
*"              RETURN STRUCTURE  BAPIRET2
*"----------------------------------------------------------------------
  CLEAR: RETURN.
* Clear message table and set bapi flag
  REFRESH G_RETURN_TB.
* CLEAR ALL WORKING STORAGE.
  PERFORM CLEAR_INFORMATION.

* Bring bapi passed variables into program structure
  MOVE-CORRESPONDING BAPIOIUDNHEADER TO ROIU_PR_DN.
  MOVE-CORRESPONDING BAPIOIUDN1DETAIL TO ROIU_PR_DN.
  ROIU_PR_DN-BUKRS = BAPIOIUDN1DETAIL-COMP_CODE.
  ROIU_PR_DN-VNAME = BAPIOIUDN1DETAIL-VENTURE.

  W_MODE = C_CREATE.
  G_BAPI_FG = C_TRUE.

* Authority check
  perform check_bapi_authorization using 'O3U_DN1'.

* changes the description texts to uppercase
  PERFORM TRANSLATE_TEXTS.
  REFRESH G_RETURN_TB.
  CLEAR G_BAPI_STOP.
* Get number range and begin creation of delivery network
  IF ROIU_PR_DN-DN_GRP IS INITIAL.
    PERFORM CHECK_SINGLE_GRP.
  ENDIF.
  perform CHECK_FOR_NUMBER.
  PERFORM NUMBER_RANGE_CHECK_BAPI.
* Make bapi edit checks.
  PERFORM BAPI_EDIT_CHECKS.
* Return all messages from the edit checks and exit the function module
  CHECK_BAPI_STOP.
*  IF g_bapi_stop = c_true.
*    return[] = g_return_tb[].
*    EXIT.
*  ENDIF.
* Check for create by reference and get reference fields if necessary
*  IF NOT ROIU_PR_DN-REF_DN_NO IS INITIAL.
*    W_REF_FG = C_FLAG_ON.
*    PERFORM GET_REF_DELIVERY_NETWORK USING ROIU_PR_DN-REF_DN_NO
*                                           TEXT-001
*                                           TEXT-002
*                                           ROIU_PR_DN-REF_DN_NO
*                                           SPACE.
*  ENDIF.
```

```
*
* Validate that the set id passed is valid for the standard hierarchy.
  PERFORM CHECK_SETID USING ROIU_PR_DN-DN_SET.
* Get unique delivery network number set up internal table data
* and insert data into db table
  PERFORM GET_DEL_NETWORK_NUMBER.
* Check for DN number in Razorfish table                  "N 1736401
  PERFORM check_razorfish_dn_duplicate.                   "N 1736401
* Lock the delivery network unique number once you get it
  PERFORM LOCK_DELIVERY_NUMBER.
* Enqueue DN Standard Hierarchy                           " SOGK013053 EY
  PERFORM enqueue_setid USING space ROIU_PR_DN-DN_SET.    " SOGK013053 EY
  CHECK_BAPI_STOP.                                        " SOGK013053 EY
* Set all data in internal tables before the save
  PERFORM PREPARE_CREATE_FOR_SAVE.
* Calls the funuction module to insert the new record into the DB
  PERFORM SAVE_DEL_NET_DATA.
  Perform F_UPDATE_HIERARCHY.                          "SOGK027671 VV
* Return error message from the save routine and exits the function mod
   CHECK_BAPI_STOP.
*  IF g_bapi_stop = c_true.
*    return[] = g_return_tb[].
*    EXIT.
*  ENDIF.
* Sends user a success message
  PERFORM SAVE_SUCCESSFUL.
* Return success message from the save and exits the function module
  RETURN[] = G_RETURN_TB[].

* Move the new records to the return structure
  MOVE-CORRESPONDING ROIU_PR_DN TO BAPIOIUDNHEADERE.
  MOVE-CORRESPONDING ROIU_PR_DN TO BAPIOIUDN1DETAILE.

ENDFUNCTION.
```

Subroutines:		Line	Page
BAPI_EDIT_CHECKS.....	LOIU_DELIVERY_NETWORKU02	46	4
CHECK_BAPI_AUTHORIZATION.....	LOIU_DELIVERY_NETWORKU02	33	4
CHECK_FOR_NUMBER.....	LOIU_DELIVERY_NETWORKU02	43	4
CHECK_RAZORFISH_DN_DUPLICATE..	LOIU_DELIVERY_NETWORKU02	69	5
CHECK_SETID.....	LOIU_DELIVERY_NETWORKU02	64	5
CHECK_SINGLE_GRP.....	LOIU_DELIVERY_NETWORKU02	41	4
CLEAR_INFORMATION.....	LOIU_DELIVERY_NETWORKU02	21	4
ENQUEUE_SETID.....	LOIU_DELIVERY_NETWORKU02	73	5
F_UPDATE_HIERARCHY.....	LOIU_DELIVERY_NETWORKU02	79	5
GET_DEL_NETWORK_NUMBER.....	LOIU_DELIVERY_NETWORKU02	67	5
LOCK_DELIVERY_NUMBER.....	LOIU_DELIVERY_NETWORKU02	71	5
NUMBER_RANGE_CHECK_BAPI.....	LOIU_DELIVERY_NETWORKU02	44	4
PREPARE_CREATE_FOR_SAVE.....	LOIU_DELIVERY_NETWORKU02	76	5
SAVE_DEL_NET_DATA.....	LOIU_DELIVERY_NETWORKU02	78	5
SAVE_SUCCESSFUL.....	LOIU_DELIVERY_NETWORKU02	87	5
TRANSLATE_TEXTS.....	LOIU_DELIVERY_NETWORKU02	36	4

```
FUNCTION BAPI_DELIVERY_NETWORK_READ.
*"----------------------------------------------------------------------
*"*"Local interface:
*"       IMPORTING
*"             VALUE(BAPIDNNUMBER) LIKE  BAPI_OIU_DN-DN_NO
*"       EXPORTING
*"             VALUE(BAPIOIUDNHEADERE) LIKE  BAPI_OIU_DN
*"                                 STRUCTURE  BAPI_OIU_DN
*"             VALUE(BAPIOIUDN1DETAILE) LIKE  BAPI_OIU_DN_1
*"                                 STRUCTURE  BAPI_OIU_DN_1
*"       TABLES
*"              RETURN STRUCTURE  BAPIRET2
*"----------------------------------------------------------------------
* Clear message table and set bapi flag
  REFRESH G_RETURN_TB.
* CLEAR ALL WORKING STORAGE.
  PERFORM CLEAR_INFORMATION.

  G_BAPI_FG = C_TRUE.

* Authority check
  perform check_bapi_authorization using 'O3U_DN3'.

  ROIU_PR_DN-DN_NO = BAPIDNNUMBER.

* Get the delivery network detail information
  PERFORM GET_REF_DELIVERY_NETWORK USING ROIU_PR_DN-DN_NO
                                         TEXT-002
                                         ROIU_PR_DN-DN_NO
                                         SPACE
                                         SPACE.

  IF G_BAPI_STOP = C_TRUE.
    RETURN[] = G_RETURN_TB[].
    EXIT.
  ELSE.
    MOVE-CORRESPONDING ROIU_PR_DN TO BAPIOIUDNHEADERE.
    MOVE-CORRESPONDING ROIU_PR_DN TO BAPIOIUDN1DETAILE.
  ENDIF.




ENDFUNCTION.
```

Subroutines:		Line	Page
CHECK_BAPI_AUTHORIZATION.....	LOIU_DELIVERY_NETWORKU03	22	7
CLEAR_INFORMATION.....	LOIU_DELIVERY_NETWORKU03	17	7
GET_REF_DELIVERY_NETWORK.....	LOIU_DELIVERY_NETWORKU03	27	7

```
FUNCTION IDOC_OUTPUT_OIU_DELIVERY_NETWK.
*"————————————————————————————————————————————————————————————————————————
*"*"Update function module:
*"
*"*"Local interface:
*"  IMPORTING
*"     VALUE(I_ROIU_PR_DN) LIKE  ROIU_PR_DN STRUCTURE  ROIU_PR_DN
*"     VALUE(REFRESH) TYPE  CHAR1 OPTIONAL
*"————————————————————————————————————————————————————————————————————————
DATA: W_LS LIKE BDI_LOGSYS-LOGSYS.
DATA: BEGIN OF CNTRL_DATA OCCURS 0.
       INCLUDE STRUCTURE EDP13.
DATA: END OF CNTRL_DATA.
DATA: W_MESTYP LIKE EDP13-MESTYP.
data: C_RFSH_IDOC_MESTYP LIKE EDP13-MESTYP VALUE 'OIURDN1'.

  CLEAR: CNTRL_DATA,
         IDOC_CNTRL.
REFRESH: CNTRL_DATA,
         IDOC_CNTRL.

IF REFRESH IS INITIAL.
  W_MESTYP = C_IDOC_MESTYP.
ELSE.
  W_MESTYP = C_RFSH_IDOC_MESTYP.
ENDIF.

*- set control data ————————————————————————4.6A————————————————————————*
CALL FUNCTION 'OIU_GET_LS_PARTNER_PROFILES'
     EXPORTING
          I_IDOC_MESTYP           = W_MESTYP
     IMPORTING
          I_LS                    = W_LS
     TABLES
          I_CNTRL_DATA            = CNTRL_DATA
     EXCEPTIONS
          NO_LOGICAL_SYSTEM_FOUND = 1
          NO_PARTNER_DATA_FOUND   = 2
          OTHERS                  = 3.

IF SY-SUBRC = 0.
  PERFORM FILL_CONTROL_SEGMENT TABLES IDOC_CNTRL
                                      CNTRL_DATA.
*Fill the internal table IDOC_DATA ————————————————————————————————————*
  PERFORM FILL_DATA_SEGMENT USING I_ROIU_PR_DN.
*Create the IDOC ——————————————————————————————————————————————————————*
  PERFORM SEND_IDOC.
ENDIF.
ENDFUNCTION.
```

Subroutines:		Line	Page
FILL_CONTROL_SEGMENT.....	LOIU_DELIVERY_NETWORKU06	42	9
FILL_DATA_SEGMENT.....	LOIU_DELIVERY_NETWORKU06	45	9
SEND_IDOC.....	LOIU_DELIVERY_NETWORKU06	47	9

External calls

Function modules:		Line	Page
OIU_GET_LS_PARTNER_PROFILES...	LOIU_DELIVERY_NETWORKU06	29	9

```
FUNCTION OIU_GET_DELIVERY_NETWORK.
*"----------------------------------------------------------------------
*"*"Local interface:
*"       IMPORTING
*"             VALUE(I_DN_NO) LIKE  OIU_PR_DN-DN_NO
*"       EXPORTING
*"             VALUE(E_ROIU_PR_DN) LIKE  ROIU_PR_DN
*"                                 STRUCTURE  ROIU_PR_DN
*"             VALUE(E_NOT_FOUND) LIKE  ROIU_PR_DN-XXDBACTION
*"       EXCEPTIONS
*"              NOT_FOUND
*"----------------------------------------------------------------------

************************************************************************
*              Master data select using full key                       *
************************************************************************

  SELECT SINGLE * FROM OIU_PR_DN
                  WHERE DN_NO = I_DN_NO.

  IF SY-SUBRC NE 0.
    E_NOT_FOUND = C_TRUE.
  ELSE.
    E_ROIU_PR_DN = OIU_PR_DN.
  ENDIF.

ENDFUNCTION.
```

DATABASE OPERATIONS

SELECT:		Line	Page
OIU_PR_DN.....	LOIU_DELIVERY_NETWORKU05	18	12

```
FUNCTION OIU_UPDATE_DELIVERY_NETWORK.
*"----------------------------------------------------------------------
*"*"Update function module:
*"
*"*"Local interface:
*"  IMPORTING
*"     VALUE(I_DELIVERY_NETWORK) LIKE  ROIU_PR_DN STRUCTURE  ROIU_PR_DN
*"----------------------------------------------------------------------

DATA:  BEGIN OF dn_tb OCCURS 0.
         INCLUDE STRUCTURE ROIU_PR_DN.
DATA:  END OF dn_tb.

CLEAR:  dn_tb.
REFRESH:  dn_tb.

  MOVE-CORRESPONDING I_DELIVERY_NETWORK TO OIU_PR_DN.
  CASE I_DELIVERY_NETWORK-XXDBACTION.
    WHEN 'I'.
      INSERT OIU_PR_DN.
    WHEN 'U'.
      UPDATE OIU_PR_DN.
  ENDCASE.

  IF SY-SUBRC <> 0.
     MESSAGE A899(OIUCM) WITH text-035.
   ENDIF.

*Razorfish Integration starts here

MOVE-CORRESPONDING i_delivery_network to dn_tb.
APPEND dn_tb.

   CALL FUNCTION 'OIUHS_INTG_PR_DN35'
        TABLES
             IN_PR_DN = DN_TB
             .

*Razorfish Integration ends here

ENDFUNCTION.
```

Messages:

		Line	Page
OIUCM899.....	LOIU_DELIVERY_NETWORKU01	26	14

External calls

Function modules:		Line	Page
OIUHS_INTG_PR_DN35.....	LOIU_DELIVERY_NETWORKU01	34	14

DATABASE OPERATIONS

UPDATE:		Line	Page
OIU_PR_DN.....	LOIU_DELIVERY_NETWORKU01	22	14

```
FUNCTION-POOL OIU_DELIVERY_NETWORK          MESSAGE-ID OIUCM.
**********************************************************************
* Project        : E&P (Exploration and Production)
* Program Name   : SAPLOIU_DELIVERY_NETWORK
* Program Title  : Delivery Network Identification
* Date Written   : 06/02/99
* Author         : Gregory A. Kisver
* Description    : (Detailed description of the program and its purpose)
* Tables Updated : List the tables actually updated by this program
* Special Logic  : Describe any special logic within the program
* Includes       : List includes and their purpose
* Return Codes   : List all return codes and description
* Input  Parameters for Includes/Function Modules:
* List the input parameters and a description
* Output Parameters for Includes/function Modules:
* List the output parameters and a description
**********************************************************************
*____________________________________________________________________*
*               C H A N G E   H I S T O R Y   B L O C K              *
*____________________________________________________________________*
*——>> Please copy the following lines of code when a change is made to *
*——>> this program and fill the change information                   *
*____________________________________________________________________*
* Change Request Number:
*  (- New lines of code: add this number to the end of the line of code.
*   - Lines to be changed: copy the line prior to changing it, comment
*     the line to be changed out and change the copied line. Add this
*     number to both lines of code.
*   - Lines to be deleted: Comment these lines out and add this number
*     to the lines of code.)
* Change date: MM/DD/YYYY
* Changed by : (Name of person making changes)
* Description: (Detailed description of change)
*____________________________________________________________________*
TABLES:  T005T,
         T001,
         T8JVT,
         T8JV,
         DD07V,
         oiu_do_jv,
         OIU_CM_LXPGL_TX,
         OIU_CM_LXSGL_TX,
         OIU_PR_DN_GRP_TX,
         OIU_CM_TXN_CTL,
         OIU_CM_EDIT,
         OIU_PR_DN_GRP,
         ROIU_PR_DN,
         ROIU_ADMINISTRATIVE,
         SETHEADERT,
         EDP13,                    "IDOC RFC CONTROL TABLE
         E1OIUDN.                  "IDOC DELIVERY NETWORK SEGMENT


*____________________________________________________________________*
*                       Working Variables                            *
*____________________________________________________________________*
DATA:        W_FCODE       LIKE  SY-UCOMM,
             W_MODE        LIKE  OIU_CM_TXN_CTL-MCODE,
             W_NEW_MODE    LIKE  W_MODE,
             W_TTLTXT      LIKE  SY-MSGV1,
             W_REF_FG      TYPE C,                   "Create by ref flag
```

```
              W_ANSWER      TYPE C,
              W_LONG_TEXT_READ_FG          TYPE C,
              W_LONG_TEXT_FG               TYPE C,
              W_TEMP_LONG_TEXT_WRITTEN_FG  TYPE C,
              W_LONG_TEXT_WRITTEN_FG       TYPE C,
              W_DOMAIN_VALUE LIKE DD07V-DOMVALUE_L,
              W_DDTEXT       LIKE DD07V-DDTEXT,
              W_GEO_REQUIRED_FG   TYPE C,
              W_SEC_REQUIRED_FG   TYPE C,
              W_CNTRY             LIKE OIU_PR_DN-CNTRY_CD,
              W_PRIMARY           LIKE OIU_PR_DN-PRI_GEO_LOC,
              OK_CODE             LIKE SY-UCOMM,
              W_NO_IDOC           TYPE C,
              W_PROGRAM           LIKE SY-CPROG,
              W_DYNNR             LIKE SY-DYNNR,
              W_DISPLAY(1)        TYPE C,
              W_OLD_SET           LIKE SETHEADER-SETNAME,
              W_NEW_SET           LIKE SETHEADER-SETNAME,
              W_SET_OBJECT        LIKE SETVALUES-FROM.
*                                                        Start SOGK013053 EY
  Data:
        w_1st_time(1)             type c,
        w_HIER_ACTIVE             like OIU_CM_PARAM-EP_ACTIVE.

Define m_HIER_ACTIVE.
* Get the E&P set parameters —— once
  If w_1st_time is initial.
     w_1st_time  = c_yes.

     SELECT SINGLE
            HIER_active
       INTO w_HIER_ACTIVE
       FROM oiu_cm_param
     CLIENT SPECIFIED
      WHERE mandt = sy-mandt.
  EndIf.
END-OF-DEFINITION.
*                                                        Start SOGK013053 EY
*————————————————————————————————————————————————————————————————————————*
*                          CONSTANTS                                     *
*————————————————————————————————————————————————————————————————————————*
CONSTANTS:    C_OFF_FG      TYPE C          VALUE 0,      "Screen Off Flag
              C_gas_plant_jv   like oiu_do_jv-vname value '04',
              C_FLAG_ON     TYPE C          VALUE 'X',    "Flag On
              C_FLAG_OFF    TYPE C          VALUE ' ',    "Flag Off
              C_YES         TYPE C          VALUE 'J',    "Yes answer
              C_UPDATE_ACT  TYPE C          VALUE 'U',    "DB Update
              C_INSERT_ACT  TYPE C          VALUE 'I',    "DB Insert
              C_CREATE(2)   TYPE C          VALUE '01',     "Create Mode
              C_UPDATE(2)   TYPE C          VALUE '02',     "Update Mode
              C_DISPLAY(2)  TYPE C          VALUE '03',    "Display Mode
              C_ENTER       LIKE SY-UCOMM   VALUE 'ENTR', "Enter fcode
              C_SAVE        LIKE SY-UCOMM   VALUE 'SAVE', "Save  fcode
              C_BACK        LIKE SY-UCOMM   VALUE 'BACK', "Back  fcode
              C_EXIT        LIKE SY-UCOMM   VALUE 'EXIT', "Exit  fcode
              C_ADMN        LIKE SY-UCOMM   VALUE 'ADMN', "Admin Data fc
              C_OBJC        LIKE SY-UCOMM   VALUE 'OBJC', "object changes
              C_OTHR        LIKE SY-UCOMM   VALUE 'OTHR', "new fcode
              C_MESSAGEID_CM  LIKE SY-MSGID VALUE 'OIUCM', "msg class CM
              C_MESSAGEID_PR  LIKE SY-MSGID VALUE 'OIUPR', "msg class PR
              C_MESSAGETYPE_S LIKE SY-MSGTY VALUE 'S',     "success Msg
```

```
              C_MESSAGETYPE_W LIKE SY-MSGTY VALUE 'W',     "Warning Msg
              C_MESSAGETYPE_E LIKE SY-MSGTY VALUE 'E',     "Error Msg
*              C_NUM_RNG_OBJ   LIKE INRI-OBJECT VALUE 'OIU_DN',
              C_INTERNAL(11)  TYPE C        VALUE 'INTERN-0001',
              C_DUMMY         LIKE SY-MSGV1 VALUE SPACE,   "Dummy funct
              C_GAS_PLANT     TYPE C        VALUE 'P',
              C_FUEL_SYSTEM   TYPE C        VALUE 'F',
              C_CLUSTER       TYPE C        VALUE 'C',
              C_GATHERING_SYS TYPE C        VALUE 'N',
              C_US            LIKE OIU_CM_LXPGL-PRI_GEO_LOC VALUE 'US',
              C_CANADA        LIKE C_US     VALUE 'CA',
              C_LONG          LIKE SY-UCOMM VALUE 'LONG',
              C_LONGTXT(7)    TYPE C        VALUE 'LONGTXT',
              C_LTXT          LIKE THEAD-TDID VALUE 'LTXT',
              C_DN_TAB_NAME(9) TYPE C       VALUE 'OIU_PR_DN',
              C_DN_TYPECD_DOM  LIKE DD07V-DOMNAME VALUE 'OIU_DN_TYP_CD',
              C_ZERO            TYPE C     VALUE '0',
              C_LT_THOUSAND(3)  TYPE C     VALUE '999',
              C_CANCEL          TYPE C     VALUE 'A',
              C_TYPE_CD         LIKE OIU_PR_DN-TYPE_CD VALUE 'N',
              C_FCODE_DND1      LIKE SY-UCOMM    VALUE 'DND1',
              C_FCODE_DND2      LIKE SY-UCOMM    VALUE 'DND2',
              C_FCODE_DND3      LIKE SY-UCOMM    VALUE 'DND3',
              C_FCODE_DND6      LIKE SY-UCOMM    VALUE 'DND6',
              C_FCODE_LOC       LIKE SY-UCOMM    VALUE 'LOC',
              C_TRANS_DND1      LIKE SY-UCOMM    VALUE 'O3U_DND1',
              C_TRANS_DND2      LIKE SY-UCOMM    VALUE 'O3U_DND2',
              C_TRANS_DND3      LIKE SY-UCOMM    VALUE 'O3U_DND3',
              C_TRANS_DND6      LIKE SY-UCOMM    VALUE 'O3U_DND6',
              C_SCREEN_1100     LIKE SY-DYNNR    VALUE '1100',
*              C_SCREEN_1200     LIKE SY-DYNNR    VALUE '1200',
              C_DN_SET_FNAME    LIKE DYNPREAD-FIELDNAME
                                VALUE 'ROIU_PR_DN-DN_SET',
              C_SETCLASS        LIKE SETHEADER-SETCLASS VALUE '8001',
*The following constants are for IDOC output creation ______________**
              C_IDOC_MESTYP     LIKE EDP13-MESTYP VALUE 'OIUDN1',
              C_IDOC_DOCTYP     LIKE EDP13-DOCTYP VALUE 'OIUDN01',
              C_IDOC_DIRECT     LIKE EDIDC-DIRECT VALUE '1'.

*____________________________________________________________________*
*                        Internal Tables                             *
*____________________________________________________________________*
DATA:  BEGIN OF DELIVERY_NETWORK_TB1 OCCURS 1.
         INCLUDE STRUCTURE ROIU_PR_DN.
DATA:  END   OF DELIVERY_NETWORK_TB1.

DATA:  BEGIN OF DELIVERY_NETWORK_TB2 OCCURS 1.
         INCLUDE STRUCTURE DELIVERY_NETWORK_TB1.
DATA:  END   OF DELIVERY_NETWORK_TB2.

DATA:  FCODE_TB  LIKE SY-UCOMM OCCURS 10 WITH HEADER LINE.
*   Reference Delivery network itab
DATA:  BEGIN OF REF_DELIVERY_NETWORK_DS.
DATA:    DN_DE          LIKE ROIU_PR_DN-DN_DE,
         DN_SET         LIKE ROIU_PR_DN-DN_SET,
         CNTRY_CD       LIKE ROIU_PR_DN-CNTRY_CD,
         PRI_GEO_LOC    LIKE ROIU_PR_DN-PRI_GEO_LOC,
         SEC_GEO_LOC    LIKE ROIU_PR_DN-SEC_GEO_LOC,
         TOL_PC         LIKE ROIU_PR_DN-TOL_PC,
         BUKRS          LIKE ROIU_PR_DN-BUKRS,
         VNAME          LIKE ROIU_PR_DN-VNAME,
```

```
          GPLT_DN_NO     LIKE ROIU_PR_DN-GPLT_DN_NO.
DATA:  END  OF REF_DELIVERY_NETWORK_DS.
*THE FOLLOWING INTERNAL TABLE IS FOR IDOC CREATION
DATA:  BEGIN OF IDOC_EDP13 OCCURS 1.
          INCLUDE STRUCTURE EDP13.
DATA:  END   OF IDOC_EDP13.

DATA:  BEGIN OF IDOC_CNTRL OCCURS 1.
          INCLUDE STRUCTURE EDIDC.
DATA:  END   OF IDOC_CNTRL.

DATA: BEGIN OF IDOC_DATA OCCURS 0.
         INCLUDE STRUCTURE EDIDD.
DATA: END OF IDOC_DATA.
DATA: W_FORMAT.
*———————————————————————————————————————————————————————————————————*
*                       Data Structures                               *
*———————————————————————————————————————————————————————————————————*
DATA: W_RETURN_DS    LIKE BAPIRET2.
DATA: W_INTERVAL_DS  LIKE NRIV.
*———————————————————————————————————————————————————————————————————*
*                       Include Programs                              *
*———————————————————————————————————————————————————————————————————*
* Change document include
INCLUDE: FOIU_DNCDT,
         FOIU_DNCDC,
         LOIU_COMMONTP2,
         LOIU_COMMONF02.



* Changed by VV on 20020729 for Maintaining Production Number range

DATA:  wa_oiu_cm_edit TYPE oiu_cm_edit.
DATA:  g_tabname TYPE oiu_cm_sn_nr-tabname.


LOAD-OF-PROGRAM.
  SELECT SINGLE *
    FROM oiu_cm_edit
    INTO wa_oiu_cm_edit
    WHERE edit_no = '0007'.
  CHECK sy-subrc EQ 0.
* check to see if integration is on or not
  IF NOT wa_oiu_cm_edit-status IS INITIAL.
* integration is turned on
    g_tabname = 'OIU_PR_DN'.
  ELSE.
* integration is turned off
    g_tabname = 'OIU_PR_DN_10'.
  ENDIF.

```

Subroutines:

		Line	Page
CD_CALL_OIU_DN.....	FOIU_DNCDC	1	
CHECK_BAPI_AUTHORIZATION.....	LOIU_COMMONF02	24	
GET_SA_DT.....	LOIU_COMMONF02	122	
SET_RETURN_MESSAGE.....	LOIU_COMMONF02	33	
	LOIU_COMMONF02	47	

Messages:

		Line	Page
?.....	LOIU_COMMONF02	104	

External calls

Function modules:		Line	Page
BALW_BAPIRETURN_GET2.....	LOIU_COMMONF02	78	
OIL_LAST_DAY_OF_MONTH.....	LOIU_COMMONF02	147	
OIU_DN_WRITE_DOCUMENT.....	FOIU_DNCDC	5	

DATABASE OPERATIONS

SELECT:		Line	Page
OIU_CM_EDIT.....	LOIU_DELIVERY_NETWORKTOP	223	21

```
*———————————————————————————————————————————————————————————————*
***INCLUDE LOIU_DELIVERY_NETWORKF01 .
*———————————————————————————————————————————————————————————————*

*&———————————————————————————————————————————————————————————————*
*&      Form  GET_REF_DELIVERY_NETWORK
*&———————————————————————————————————————————————————————————————*
*       text
*———————————————————————————————————————————————————————————————*
*      ——>P_W_MESSAGE_DS  text
*———————————————————————————————————————————————————————————————*
FORM get_ref_delivery_network USING delivery_network
                                    text1
                                    text2
                                    text3
                                    w_ref_fg.

  DATA:  temp_dn_tb  LIKE oiu_pr_dn,
         not_found   TYPE c.
* Get the reference delivery network for creation

***********************************************************************
*              Select master data using full key                      *
***********************************************************************
  CALL FUNCTION 'OIU_DN_MASTER_READ'
       EXPORTING
            dn_no       = delivery_network
            force_db    = 'X'
       IMPORTING
            e_oiu_pr_dn = temp_dn_tb
       EXCEPTIONS
            none_found  = 1
            OTHERS      = 2.

  IF sy-subrc NE 0.
    PERFORM set_return_message USING c_messagetype_e
                                     c_messageid_cm
                                     '002'
                                     space
                                     space
                                     text1
                                     text2
                                     text3
                                     space
                                     space
                                     space
                                     space.
  ENDIF.

  IF w_ref_fg = c_flag_on.
    MOVE-CORRESPONDING temp_dn_tb TO ref_delivery_network_ds.
    MOVE-CORRESPONDING ref_delivery_network_ds TO roiu_pr_dn.
  ELSE.
    MOVE-CORRESPONDING temp_dn_tb TO roiu_pr_dn.
    delivery_network_tb1 = roiu_pr_dn.
  ENDIF.

**PERFORM CHECK_SETID USING ROIU_PR_DN-DN_SET.  "SO7K011423 BT
  PERFORM get_setid_desc.

ENDFORM.                    " GET_REF_DELIVERY_NETWORK
```

```
*&---------------------------------------------------------------------*
*&      Form  SET_ERROR_MESSAGE
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*      -->P_W_MESSAGE_DS  text
*----------------------------------------------------------------------*
*FORM set_error_message USING    p_message  LIKE bapiret2.
*
*  IF p_message-type = c_messagetype_e.
*    MESSAGE ID p_message-id
*    TYPE       p_message-type
*    NUMBER     p_message-number
*    WITH       p_message-message_v1
*               p_message-message_v2
*               p_message-message_v3
*               p_message-message_v4.
*  ENDIF.
*
*ENDFORM.                    " SET_ERROR_MESSAGE
*&---------------------------------------------------------------------*
*&      Form  SAVE_DEL_NET_DATA
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*      -->P_W_RETURN_DS  text
*----------------------------------------------------------------------*
FORM save_del_net_data.

  CALL FUNCTION 'OIU_UPDATE_DELIVERY_NETWORK' IN UPDATE TASK
       EXPORTING
            i_delivery_network = roiu_pr_dn.

  IF w_mode = c_create.
*    create node table entry
    CALL FUNCTION 'OIU_NODE_DN_INSERT' IN UPDATE TASK
         EXPORTING
              i_dn_no = roiu_pr_dn-dn_no.
  ENDIF.

* update hierarchy?
*  PERFORM f_update_hierarchy. "SOGK027671 VV

*create outbound idoc
  CALL FUNCTION 'IDOC_OUTPUT_OIU_DELIVERY_NETWK' IN UPDATE TASK
       EXPORTING
            i_roiu_pr_dn = roiu_pr_dn.


ENDFORM.                    " SAVE_DEL_NET_DATA
*&---------------------------------------------------------------------*
*&      Form  PREPARE_CREATE_FOR_SAVE
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  -->  p1        text
*  <--  p2        text
*----------------------------------------------------------------------*
FORM prepare_create_for_save.

  IF w_mode = c_create.
```

```
123
124       roiu_pr_dn-shtxt = roiu_pr_dn-dn_de.
125       SET LOCALE LANGUAGE sy-langu.
126      TRANSLATE roiu_pr_dn-shtxt TO UPPER CASE.            "#EC TRANSLANG
127      SET LOCALE LANGUAGE space.
128      roiu_pr_dn-xxdbaction = 'I'.
129      roiu_pr_dn-oiu_cruser = sy-uname.
130      GET TIME STAMP FIELD roiu_pr_dn-oiu_timestamp.
131      delivery_network_tb1 = roiu_pr_dn.
132      APPEND delivery_network_tb1.
133    ENDIF.
134
135  ENDFORM.                    " PREPARE_CREATE_FOR_SAVE
136  *&---------------------------------------------------------------------*
137  *&      Form  PREPARE_UPDATE_FOR_SAVE
138  *&---------------------------------------------------------------------*
139  *       text
140  *---------------------------------------------------------------------*
141  *  ---->  p1        text
142  *  <----  p2        text
143  *---------------------------------------------------------------------*
144  FORM prepare_update_for_save.
145
146    roiu_pr_dn-xxdbaction = c_update_act.                  "SOGK026707 BT
147    roiu_pr_dn-shtxt = roiu_pr_dn-dn_de.
148    SET LOCALE LANGUAGE sy-langu.
149    TRANSLATE roiu_pr_dn-shtxt  TO UPPER CASE.             "#EC TRANSLANG
150    SET LOCALE LANGUAGE space.
151
152
153  ENDFORM.                    " PREPARE_UPDATE_FOR_SAVE
154  *&---------------------------------------------------------------------*
155  *&      Form  CHECK_FOR_CHANGES
156  *&---------------------------------------------------------------------*
157  *       text
158  *---------------------------------------------------------------------*
159  *  ---->  p1        text
160  *  <----  p2        text
161  *---------------------------------------------------------------------*
162  FORM check_for_changes.
163
164
165
166
167    IF w_mode NE c_display.
168      IF delivery_network_tb1 NE roiu_pr_dn OR w_mode = c_create
169         OR w_long_text_written_fg NE space.
170        IF w_fcode = c_back.
171          PERFORM popup_confirm_step USING text-003 text-004 text-005.
172        ELSE.
173          PERFORM popup_confirm_step USING text-003 text-004 text-006.
174        ENDIF.
175      ENDIF.
176    ENDIF.
177
178  ENDFORM.                    " CHECK_FOR_CHANGES
179  *&---------------------------------------------------------------------*
180  *&      Form  POPUP_CONFIRM_STEP
181  *&---------------------------------------------------------------------*
182  *       text
183  *---------------------------------------------------------------------*
```

```
*      ——>P_TEXT_003  text
*      ——>P_TEXT_004  text
*      ——>P_TEXT_005  text
*———————————————————————————————————————————————————————————————————*
FORM popup_confirm_step USING    text1
                                 text2
                                 title.

  CALL FUNCTION 'POPUP_TO_CONFIRM_STEP'
       EXPORTING
            defaultoption  = 'Y'
            textline1      = text1
            textline2      = text2
            titel          = title
            start_column   = 15
            start_row      = 3
            cancel_display = 'X'
       IMPORTING
            answer         = w_answer.

  IF w_answer = c_yes.
    CASE w_mode.
      WHEN c_create.
        PERFORM create_delivery_network.
      WHEN c_update.
        PERFORM update_delivery_network.
    ENDCASE.
  ELSE.
    PERFORM rollback_longtext.
    PERFORM unlock_delivery_number.
  ENDIF.

ENDFORM.                    " POPUP_CONFIRM_STEP
*&———————————————————————————————————————————————————————————————————*
*&      Form  CHANGE_DOCUMENT
*&———————————————————————————————————————————————————————————————————*
*       text
*———————————————————————————————————————————————————————————————————*
*  ——>  p1        text
*  <——  p2        text
*———————————————————————————————————————————————————————————————————*
FORM change_document USING dn1   STRUCTURE roiu_pr_dn
                           dn2   STRUCTURE roiu_pr_dn.

  utime    = sy-uzeit.
  udate    = sy-datum.
  username = sy-uname.
  tcode    = sy-tcode.
  upd_oiu_pr_dn = c_update_act.
  objectid = roiu_pr_dn-dn_no.

  MOVE-CORRESPONDING dn1 TO *oiu_pr_dn.
  MOVE-CORRESPONDING dn2 TO oiu_pr_dn.

  PERFORM cd_call_oiu_dn.


ENDFORM.                    " CHANGE_DOCUMENT
*&———————————————————————————————————————————————————————————————————*
*&      Form  NO_CHANGES_MADE
*&———————————————————————————————————————————————————————————————————*
```

```
*       text
*———————————————————————————————————————————————————————————————————*
*  ——>  p1        text
*  <——  p2        text
*———————————————————————————————————————————————————————————————————*
FORM no_changes_made.

  PERFORM set_return_message USING c_messagetype_s
                                   c_messageid_cm
                                   '011'
                                   space
                                   space
                                   space
                                   space
                                   space
                                   space
                                   space
                                   space
  LEAVE TO TRANSACTION sy-tcode.

ENDFORM.                    " NO_CHANGES_MADE
*&———————————————————————————————————————————————————————————————————*
*&      Form  CHECK_FOR_NUMBER
*&———————————————————————————————————————————————————————————————————*
*       text
*———————————————————————————————————————————————————————————————————*
*  ——>  p1        text
*  <——  p2        text
*———————————————————————————————————————————————————————————————————*
FORM check_for_number.

***********************************************************************
*              Select master data using full key                      *
***********************************************************************

ENDFORM.                    " CHECK_FOR_NUMBER
*&———————————————————————————————————————————————————————————————————*
*&      Form  GET_NUMBER_RANGE_INFO
*&———————————————————————————————————————————————————————————————————*
*       text
*———————————————————————————————————————————————————————————————————*
*      ——>P_W_RETURN_DS  text
*———————————————————————————————————————————————————————————————————*
FORM get_number_range_info.

  DATA:  l_subobject TYPE nrsobj.

* see if it was read last time
  CHECK oiu_pr_dn_grp-dn_grp <> roiu_pr_dn-dn_grp.

  SELECT SINGLE * FROM oiu_pr_dn_grp WHERE dn_grp = roiu_pr_dn-dn_grp.
  IF sy-subrc <> 0.
    PERFORM set_return_message USING c_messagetype_e
                                     c_messageid_cm
                                     '002'
                                     space
                                     space
                                     text-002
                                     roiu_pr_dn-dn_grp
                                     space
```

```
                                        space
                                        space
                                        space
                                        space.
  ENDIF.

*  CALL FUNCTION 'NUMBER_GET_INFO'
*       EXPORTING
*            nr_range_nr        = oiu_pr_dn_grp-dn_nrnr
*            object             = c_num_rng_obj
**         SUBOBJECT          = ' '
**         TOYEAR             = '0000'
*       IMPORTING
*            interval           = w_interval_ds
*       EXCEPTIONS
*            interval_not_found = 1
*            object_not_found   = 2
*            OTHERS             = 3.

  l_subobject = oiu_pr_dn_grp-dn_grp.

  CALL FUNCTION 'OIU_NUMBER_GET_INFO'
       EXPORTING
            tabname            = g_tabname
            externind          = oiu_pr_dn_grp-externind
            subobject          = l_subobject
       IMPORTING
            nriv               = w_interval_ds
       EXCEPTIONS
            table_not_found    = 1
            interval_not_found = 2
            object_not_found   = 3
            other_error        = 4
            OTHERS             = 5.
  IF sy-subrc NE 0.
    PERFORM set_return_message USING sy-msgty
                                     sy-msgid
                                     sy-msgno
                                     space
                                     space
                                     sy-msgv1
                                     sy-msgv2
                                     sy-msgv3
                                     sy-msgv4
                                     space
                                     space
                                     space.
  ENDIF.

ENDFORM.                    " GET_NUMBER_RANGE_INFO
*&---------------------------------------------------------------------*
*&      Form  CHECK_FOR_VALIDITY
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*      -->P_W_RETURN_DS  text
*----------------------------------------------------------------------*
FORM check_for_validity.

  DATA:    del_number TYPE i,
           l_dn_grp_tx LIKE oiu_pr_dn_grp_tx-dn_grp_tx.
```

```
  CONSTANTS:   c_internal(2)   TYPE c    VALUE '02'.

  IF w_interval_ds-externind IS INITIAL.
    CHECK w_interval_ds-nrrangenr = c_internal.
  ENDIF.
  PERFORM get_dn_grp_tx USING roiu_pr_dn-dn_grp
                              l_dn_grp_tx.
* internal number assignments
  IF w_interval_ds-externind IS INITIAL
    AND  NOT roiu_pr_dn-dn_no IS INITIAL.
    PERFORM set_return_message USING c_messagetype_e
                                     'OIUPR'
                                     '274'
                                     space
                                     space
                                     l_dn_grp_tx
                                     space
                                     space
                                     space
                                     space
                                     space
                                     space.
* external number assignments


*—— begin change note 196949 05/10/2001 lk
*  ELSEIF W_INTERVAL_DS-EXTERNIND = C_TRUE
*      and ROIU_PR_DN-DN_NO IS INITIAL.
  ELSEIF w_interval_ds-externind = c_true.
    IF   roiu_pr_dn-dn_no IS INITIAL
      OR ( ( w_interval_ds-fromnumber GT roiu_pr_dn-dn_no )"#EC PORTABLE
      OR   ( w_interval_ds-tonumber   LT   roiu_pr_dn-dn_no ) ).
*—— end change note 196949 05/10/2001 lk

*—— begin change note 196947 05/07/2001 lk
*—— the table should not be locked for all the networks
*—— when creating a external delivery network.
      PERFORM unlock_delivery_number.
*—— end change note 196947 05/07/2001 lk

*    PERFORM SET_RETURN_MESSAGE USING C_MESSAGETYPE_E
*                                     C_MESSAGEID_CM
*                                     '017'
*                                     SPACE
*                                     SPACE
*                                     TEXT-002
*                                     L_DN_GRP_TX
*                                     TEXT-002
*                                     SPACE
*                                     SPACE
*                                     SPACE
*                                     SPACE.
      PERFORM set_return_message USING c_messagetype_e
                                       c_messageid_cm
                                       '019'
                                       space
                                       space
                                       w_interval_ds-fromnumber
                                       w_interval_ds-tonumber
                                       space
                                       space
```

```
                                          space
                                          space
                                          space.
     ENDIF.
* not in range
*  ELSEIF  W_INTERVAL_DS-EXTERNIND = C_TRUE
*    AND  ( ( W_INTERVAL_DS-FROMNUMBER GT ROIU_PR_DN-DN_NO )"#EC
*PORTABLE
*    OR   ( W_INTERVAL_DS-TONUMBER   LT   ROIU_PR_DN-DN_NO ) ).
*    PERFORM SET_RETURN_MESSAGE USING C_MESSAGETYPE_E
*                                     C_MESSAGEID_CM
*                                     '019'
*                                     SPACE
*                                     SPACE
*                                     W_INTERVAL_DS-FROMNUMBER
*                                     W_INTERVAL_DS-TONUMBER
*                                     SPACE
*                                     SPACE
*                                     SPACE
*                                     SPACE
*                                     SPACE.
*—— end change note 196949 05/10/2001 lk

  ENDIF.


ENDFORM.                    " CHECK_FOR_VALIDITY
*&———————————————————————————————————————————————————————————————*
*&      Form  GET_DEL_NETWORK_NUMBER
*&———————————————————————————————————————————————————————————————*
*       text
*———————————————————————————————————————————————————————————————*
*      ——>P_W_RETURN_DS  text
*———————————————————————————————————————————————————————————————*
FORM get_del_network_number.

* Changed by VV on 20020729 for Maintaining Production Number range

  DATA:  l_subobject TYPE nrsobj.

  IF w_interval_ds-externind IS INITIAL.
*    CALL FUNCTION 'NUMBER_GET_NEXT'
*         EXPORTING
*              nr_range_nr             = w_interval_ds-nrrangenr
*              object                  = w_interval_ds-object
*              quantity                = '1'
**             SUBOBJECT               = ' '
**             TOYEAR                  = '0000'
**             IGNORE_BUFFER           = ' '
*         IMPORTING
*              number                  = roiu_pr_dn-dn_no
**             QUANTITY                =
**             RETURNCODE              =
*         EXCEPTIONS
*              interval_not_found      = 1
*              number_range_not_intern = 2
*              object_not_found        = 3
*              quantity_is_0           = 4
*              quantity_is_not_1       = 5
*              interval_overflow       = 6
*              OTHERS                  = 7.
```

```
*

    l_subobject = roiu_pr_dn-dn_grp.
    CALL FUNCTION 'OIU_GET_NEXT_SYS_NO'
         EXPORTING
              tabname                 = g_tabname
              subobject               = l_subobject
         IMPORTING
              sys_no                  = roiu_pr_dn-dn_no
         EXCEPTIONS
              table_not_found         = 1
              object_not_found        = 2
              interval_not_found      = 3
              number_range_not_intern = 4
              interval_overflow       = 5
              OTHERS                  = 6.


    IF sy-subrc NE 0.
      PERFORM set_return_message USING sy-msgty
                                       sy-msgid
                                       sy-msgno
                                       space
                                       space
                                       sy-msgv1
                                       sy-msgv2
                                       sy-msgv3
                                       sy-msgv4
                                       space
                                       space
                                       space.
    ENDIF.
    SET PARAMETER ID 'OIU_DN' FIELD roiu_pr_dn-dn_no.
  ENDIF.


ENDFORM.                    " GET_DEL_NETWORK_NUMBER
*&---------------------------------------------------------------------*
*&      Form  CHECK_FOR_DUPLICATE
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*      -->P_W_RETURN_DS  text
*----------------------------------------------------------------------*
FORM check_for_duplicate.

  DATA:   not_found  TYPE c.
*   If in create mode and using an external number range check to see
*   if a delivery network with that number already exists
  IF w_mode = c_create AND NOT roiu_pr_dn-dn_no IS INITIAL.
*********************************************************************
*              Select master data using full key                    *
*********************************************************************
    CALL FUNCTION 'OIU_DN_MASTER_READ'
         EXPORTING
              dn_no       = roiu_pr_dn-dn_no
              force_db    = 'X'
*         IMPORTING
*              E_OIU_PR_DN =
         EXCEPTIONS
              none_found  = 1
```

```
                OTHERS      = 2.

      IF sy-subrc = 0.
        PERFORM set_return_message USING c_messagetype_e
                                         c_messageid_cm
                                         '003'
                                         space
                                         space
                                         text-002
                                         roiu_pr_dn-dn_no
                                         space
                                         space
                                         space
                                         space
                                         space.
      ENDIF.
    ENDIF.

  ENDFORM.                      " CHECK_FOR_DUPLICATE
  *&---------------------------------------------------------------------*
  *&      Form  ADMINISTRATIVE_DATA
  *&---------------------------------------------------------------------*
  *       text
  *----------------------------------------------------------------------*
  *  ---->  p1        text
  *  <----  p2        text
  *----------------------------------------------------------------------*
  FORM administrative_data.

  ***********************************************************************
  *              Select data using full key                             *
  ***********************************************************************
    SELECT SINGLE dn_grp_tx FROM oiu_pr_dn_grp_tx
                            INTO roiu_administrative-fl_grp_tx
                            WHERE dn_grp = roiu_pr_dn-dn_grp
                            AND   spras  = sy-langu.

    IF sy-subrc NE 0.
      CLEAR roiu_administrative-fl_grp_tx.
    ENDIF.
    roiu_administrative-fl_grp        = roiu_pr_dn-dn_grp.
    roiu_administrative-oiu_timestamp = roiu_pr_dn-oiu_timestamp.
    roiu_administrative-cruser        = roiu_pr_dn-oiu_cruser.

    CALL FUNCTION 'OIU_CM_CREATE_GROUP_INFO'
         EXPORTING
              i_roiu_admin = roiu_administrative.

  ENDFORM.                      " ADMINISTRATIVE_DATA
  *&---------------------------------------------------------------------*
  *&      Form  CHECK_FOR_GAS_PLANT
  *&---------------------------------------------------------------------*
  *       text
  *----------------------------------------------------------------------*
  *      ---->P_W_RETURN_DS  text
  *----------------------------------------------------------------------*
  FORM check_for_gas_plant.
    DATA:  l_type(132)  TYPE c.

  * Check that only type P delivery networks contain a company code/ JV
    IF NOT roiu_pr_dn-bukrs IS INITIAL OR
```

```
      NOT roiu_pr_dn-vname IS INITIAL.
     IF roiu_pr_dn-type_cd EQ c_gathering_sys
        OR roiu_pr_dn-type_cd EQ c_fuel_system
        OR roiu_pr_dn-type_cd EQ c_cluster.
       CASE roiu_pr_dn-type_cd.
         WHEN c_gathering_sys.
           l_type = text-033.
         WHEN c_fuel_system.
           l_type = text-032.
         WHEN c_cluster.
           l_type = text-034.
       ENDCASE.
       CLEAR w_fcode.
       PERFORM set_return_message USING c_messagetype_e
                                        c_messageid_cm
                                        '004'
                                        space
                                        space
                                        text-030
                                        space
                                        space
                                        l_type
                                        space
                                        space
                                        space.
     ENDIF.
   ENDIF.

   IF NOT roiu_pr_dn-gplt_dn_no IS INITIAL
      AND roiu_pr_dn-type_cd = c_fuel_system.
     CLEAR w_fcode.
     PERFORM set_return_message USING c_messagetype_e
                                      c_messageid_cm
                                      '004'
                                      space
                                      space
                                      text-012
                                      space
                                      text-032
                                      space
                                      space
                                      space
                                      space.
   ENDIF.

 * When DN type is 'P' and company code or joint venture are blank error
   IF roiu_pr_dn-type_cd EQ c_gas_plant
     AND ( roiu_pr_dn-bukrs IS INITIAL
     OR    roiu_pr_dn-vname IS INITIAL ).
     CLEAR w_fcode.
     PERFORM set_return_message USING c_messagetype_e
                                      c_messageid_cm
                                      '009'
                                      space
                                      space
                                      text-030
                                      space
                                      space
                                      text-012
                                      space
                                      space
```

```
                                        space.
  ELSE.
    PERFORM check_joint_venture.
  ENDIF.
ENDFORM.                    " CHECK_FOR_GAS_PLANT
*&---------------------------------------------------------------------*
*&      Form  CHECK_VALID_GASPLANT_NETWORK_I
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*      -->P_W_RETURN_DS  text
*----------------------------------------------------------------------*
FORM check_valid_gasplant_ntwk_id.

  DATA:  temp_gplt_dn_tb LIKE oiu_pr_dn.


*************************************************************************
*               Select master data using full key                       *
*************************************************************************
  CALL FUNCTION 'OIU_DN_MASTER_READ'
       EXPORTING
            dn_no       = roiu_pr_dn-gplt_dn_no
            force_db    = 'X'
       IMPORTING
            e_oiu_pr_dn = temp_gplt_dn_tb
       EXCEPTIONS
            none_found  = 1
            OTHERS      = 2.

  IF sy-subrc NE 0.
    CLEAR w_fcode.
    PERFORM set_return_message USING c_messagetype_e
                                     c_messageid_cm
                                     '002'
                                     space
                                     space
                                     text-012
                                     text-002
                                     roiu_pr_dn-gplt_dn_no
                                     space
                                     space
                                     space
                                     space.
  ELSEIF roiu_pr_dn-dn_no = roiu_pr_dn-gplt_dn_no.
    CLEAR w_fcode.
    PERFORM set_return_message USING c_messagetype_e
                                     c_messageid_pr
                                     '010'
                                     space
                                     space
                                     space
                                     space
                                     space
                                     space
                                     space
                                     space
                                     space.

  ENDIF.

```

```
ENDFORM.                    " CHECK_VALID_GASPLANT_NETWORK_I
*&---------------------------------------------------------------------*
*&      Form  CHECK_COUNTRY_AND_REGION
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  ---->  p1        text
*  <----  p2        text
*----------------------------------------------------------------------*
FORM check_country_and_region.

  CALL FUNCTION 'OIU_VERIFY_PRIMARY_GEO_LOC'
       EXPORTING
            land1                = roiu_pr_dn-cntry_cd
            pri_loc_code         = roiu_pr_dn-pri_geo_loc
       IMPORTING
            pri_geo_loc_required = w_geo_required_fg
       EXCEPTIONS
            region_required      = 1
            region_not_valid     = 2
            country_not_found    = 3
*           OTHERS               = 4
          .

  IF sy-subrc <> 0.
    IF sy-msgty = 'E'.
      CLEAR w_fcode.
    ENDIF.
    PERFORM set_return_message USING sy-msgty
                                     sy-msgid
                                     sy-msgno
                                     space
                                     space
                                     sy-msgv1
                                     sy-msgv2
                                     sy-msgv3
                                     sy-msgv4
                                     space
                                     space
                                     space.
  ENDIF.

  IF w_geo_required_fg = c_true.
    CALL FUNCTION 'OIU_VERIFY_SECONDARY_GEO_LOC'
       EXPORTING
            land1                 = roiu_pr_dn-cntry_cd
            pri_loc_code          = roiu_pr_dn-pri_geo_loc
            sec_loc_code          = roiu_pr_dn-sec_geo_loc
       EXCEPTIONS
            county_city_required  = 1
            county_city_not_valid = 2
            country_not_found     = 3
            pri_geo_loc_not_found = 4
*         OTHERS                = 5
            .

    IF sy-subrc <> 0.
      IF sy-msgty = 'E'.
        CLEAR w_fcode.
      ENDIF.
      PERFORM set_return_message USING sy-msgty
```

```
                                        sy-msgid
                                        sy-msgno
                                        space
                                        space
                                        sy-msgv1
                                        sy-msgv2
                                        sy-msgv3
                                        sy-msgv4
                                        space
                                        space
                                        space.
      ENDIF.
    ENDIF.

ENDFORM.                      " CHECK_COUNTRY_AND_REGION
*&———————————————————————————————————————————————————————————————————*
*&      Form  LOCK_DELIVERY_NUMBER
*&———————————————————————————————————————————————————————————————————*
*       text
*———————————————————————————————————————————————————————————————————*
*      ——>P_W_RETURN_DS  text
*———————————————————————————————————————————————————————————————————*
FORM lock_delivery_number.

  CALL FUNCTION 'ENQUEUE_E_OIU_DN'
      EXPORTING
           mode_oiu_pr_dn = 'E'
           mandt          = sy-mandt
           dn_no          = roiu_pr_dn-dn_no
*           X_DN_NO        = ' '
*           _SCOPE         = '2'
*           _WAIT          = ' '
*           _COLLECT       = ' '
      EXCEPTIONS
           foreign_lock   = 1
           system_failure = 2
           OTHERS         = 3.

  CASE sy-subrc.
    WHEN 1.
      PERFORM set_return_message USING c_messagetype_e
                                       c_messageid_cm
                                       '025'
                                       space
                                       space
                                       text-002
                                       roiu_pr_dn-dn_no
                                       space
                                       sy-msgv1
                                       space
                                       space
                                       space.
    WHEN 2.
      PERFORM set_return_message USING c_messagetype_e
                                       c_messageid_cm
                                       '026'
                                       space
                                       space
                                       space
                                       space
                                       space
```

```
                                          space
                                          space
                                          space
                                          space.
     WHEN 3.
       PERFORM set_return_message USING c_messagetype_e
                                          c_messageid_cm
                                          '027'
                                          space
                                          space
                                          space
                                          space
                                          space
                                          space
                                          space
                                          space
                                          space.
   ENDCASE.
ENDFORM.                      " LOCK_DELIVERY_NUMBER
*&---------------------------------------------------------------------*
*&      Form  UNLOCK_DELIVERY_NUMBER
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  ---->  p1        text
*  <----  p2        text
*----------------------------------------------------------------------*
FORM unlock_delivery_number.

   CALL FUNCTION 'DEQUEUE_E_OIU_DN'
       EXPORTING
            mode_oiu_pr_dn = 'E'
            mandt          = sy-mandt
            dn_no          = roiu_pr_dn-dn_no
*         X_DN_NO        = ' '
*         _SCOPE         = '3'
*         _SYNCHRON      = ' '
*         _COLLECT       = ' '
             .
ENDFORM.                      " UNLOCK_DELIVERY_NUMBER
*&---------------------------------------------------------------------*
*&      Form  LEAVE_TO_TRANSACTION
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*      ---->P_W_RETURN_DS  text
*      ---->P_C_CREATE  text
*----------------------------------------------------------------------*
FORM leave_to_transaction USING    mode.


   CALL FUNCTION 'OIU_LEAVE_TXN_CTL'
        EXPORTING
             i_mcode    = mode
             i_cprog    = sy-cprog
             i_format   = w_format
        EXCEPTIONS
             none_found = 1
             OTHERS     = 2.


```

```
  IF sy-subrc <> 0.
    PERFORM set_return_message USING sy-msgty
                                     sy-msgid
                                     sy-msgno
                                     space
                                     space
                                     sy-msgv1
                                     sy-msgv2
                                     sy-msgv3
                                     sy-msgv4
                                     space
                                     space
                                     space.
  ENDIF.

ENDFORM.                    " LEAVE_TO_TRANSACTION
*&---------------------------------------------------------------------*
*&      Form  USE_LONG_TEXT_PROCESSING
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  ---->  p1        text
*  <----  p2        text
*----------------------------------------------------------------------*
FORM use_long_text_processing  USING  long_text_read_fg
                                      long_text_fg
                                      c_ltxt
                                      long_text_written_fg.

  DATA: long_text_name     LIKE thead-tdname.

  MOVE roiu_pr_dn-dn_no TO long_text_name(20).
  IF long_text_read_fg = c_flag_off.
    w_new_mode = c_display.
    CALL FUNCTION 'OIU_LONG_TEXT_MAINTAIN'
         EXPORTING
              oiu_flg_only_check   = c_flag_on
              oiu_modus            = w_new_mode
              oiu_table_name       = c_dn_tab_name
              oiu_text_id          = c_ltxt
              oiu_text_name        = long_text_name
*             OIU_TEXT_NAME_REF    = '              '
         IMPORTING
              oiu_flg_text_exists  = long_text_fg
              oiu_flg_txt_function = long_text_written_fg.

    long_text_read_fg = c_flag_on.
  ENDIF.

  CASE w_mode.
    WHEN c_display.
      IF long_text_fg <> c_flag_off.
        w_new_mode = c_display.
        CALL FUNCTION 'OIU_LONG_TEXT_MAINTAIN'
             EXPORTING
                  oiu_flg_only_check   = c_flag_off
                  oiu_modus            = w_new_mode
                  oiu_table_name       = c_dn_tab_name
                  oiu_text_id          = c_ltxt
                  oiu_text_name        = long_text_name
*                 oiu_text_name_ref    = '              '
```

```
977                 IMPORTING
978  *                   oiu_flg_text_exists  =
979                      oiu_flg_txt_function = long_text_written_fg.
980          ELSE.
981            MESSAGE i036.
982          ENDIF.
983        WHEN OTHERS.
984          IF long_text_fg <> c_flag_off.
985            w_new_mode = c_update.
986            CALL FUNCTION 'OIU_LONG_TEXT_MAINTAIN'
987                 EXPORTING
988                      oiu_flg_only_check   = c_flag_off
989                      oiu_modus            = w_new_mode
990                      oiu_table_name       = c_dn_tab_name
991                      oiu_text_id          = c_ltxt
992                      oiu_text_name        = long_text_name
993  *                   OIU_TEXT_NAME_REF    = '              '
994                 IMPORTING
995  *                   OIU_FLG_TEXT_EXISTS  =
996                      oiu_flg_txt_function = long_text_written_fg.
997          ELSE.
998            w_new_mode = c_create.
999            CALL FUNCTION 'OIU_LONG_TEXT_MAINTAIN'
1000                EXPORTING
1001                     oiu_flg_only_check   = c_flag_off
1002                     oiu_modus            = w_new_mode
1003                     oiu_table_name       = c_dn_tab_name
1004                     oiu_text_id          = c_ltxt
1005                     oiu_text_name        = long_text_name
1006 *                   OIU_TEXT_NAME_REF    = '              '
1007                IMPORTING
1008 *                   OIU_FLG_TEXT_EXISTS  =
1009                     oiu_flg_txt_function = long_text_written_fg.
1010         ENDIF.
1011       ENDCASE.
1012
1013 ENDFORM.                    " USE_LONG_TEXT_PROCESSING
1014 *&---------------------------------------------------------------------*
1015 *&      Form  COMMIT_ALL_WORK
1016 *&---------------------------------------------------------------------*
1017 *       text
1018 *----------------------------------------------------------------------*
1019 *  -->  p1        text
1020 *  <--  p2        text
1021 *----------------------------------------------------------------------*
1022 FORM commit_all_work.
1023
1024   CALL FUNCTION 'COMMIT_TEXT'.
1025   COMMIT WORK.
1026
1027 ENDFORM.                    " COMMIT_ALL_WORK
1028 *&---------------------------------------------------------------------*
1029 *&      Form  PREPARE_LONG_TEXT
1030 *&---------------------------------------------------------------------*
1031 *       text
1032 *----------------------------------------------------------------------*
1033 *  -->  p1        text
1034 *  <--  p2        text
1035 *----------------------------------------------------------------------*
1036 FORM prepare_long_text.
1037
```

```
*     Temporary long text change flags are used because if the call
*     function is entered more than once the value passed back to the
*     program will clear your original flag when it should be set since
*     a prior change had been made
  SET TITLEBAR c_longtxt WITH w_ttltxt.
  PERFORM use_long_text_processing USING w_long_text_read_fg
                                         w_long_text_fg
                                         c_ltxt
                                        w_temp_long_text_written_fg.
  IF NOT w_temp_long_text_written_fg IS INITIAL.
    MOVE w_temp_long_text_written_fg TO w_long_text_written_fg.
  ENDIF.

ENDFORM.                    " PREPARE_LONG_TEXT
*&---------------------------------------------------------------------*
*&      Form  GET_COUNTRY_TEXTS
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  ---->  p1        text
*  <----  p2        text
*----------------------------------------------------------------------*
FORM get_country_texts.

  IF NOT roiu_pr_dn-cntry_cd IS INITIAL.
***********************************************************************
*              Select master data using full key                      *
***********************************************************************
    IF t005t-land1 NE roiu_pr_dn-cntry_cd.
      SELECT SINGLE * FROM t005t WHERE spras = sy-langu
                                 AND   land1 = roiu_pr_dn-cntry_cd.
    ENDIF.
  ENDIF.

  IF NOT roiu_pr_dn-pri_geo_loc IS INITIAL.
***********************************************************************
*              Select master data using full key                      *
***********************************************************************
    IF oiu_cm_lxpgl_tx-pri_geo_loc NE roiu_pr_dn-pri_geo_loc.
      SELECT SINGLE * FROM oiu_cm_lxpgl_tx
                      WHERE spras       = sy-langu
                      AND   land1       = roiu_pr_dn-cntry_cd
                      AND   pri_geo_loc = roiu_pr_dn-pri_geo_loc.
    ENDIF.
  ELSE.
    CLEAR: oiu_cm_lxpgl_tx-pri_geo_desc.
  ENDIF.

  IF NOT roiu_pr_dn-sec_geo_loc IS INITIAL.
***********************************************************************
*              Select master data using full key                      *
***********************************************************************
    SELECT SINGLE * FROM oiu_cm_lxsgl_tx
                    WHERE spras       = sy-langu
                    AND   land1       = roiu_pr_dn-cntry_cd
                    AND   pri_geo_loc = roiu_pr_dn-pri_geo_loc
                    AND   sec_geo_loc = roiu_pr_dn-sec_geo_loc.

  ELSE.
    CLEAR: oiu_cm_lxsgl_tx-sec_geo_desc.
  ENDIF.
```

```
ENDFORM.                    " GET_COUNTRY_TEXTS
*&---------------------------------------------------------------------*
*&      Form  GET_TYPECD_TEXTS
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  ---->  p1        text
*  <----  p2        text
*----------------------------------------------------------------------*
FORM get_typecd_texts.

  DATA: domain_value  LIKE dd07v-domvalue_l.

  domain_value = roiu_pr_dn-type_cd.

  IF NOT roiu_pr_dn-type_cd IS INITIAL.
    CALL FUNCTION 'DOMAIN_VALUE_GET'
         EXPORTING
              i_domname  = c_dn_typecd_dom
              i_domvalue = domain_value
        IMPORTING
              e_ddtext   = dd07v-ddtext
*       EXCEPTIONS
*             NOT_EXIST  = 1
*             OTHERS     = 2
        .
  ELSE.
    CLEAR: dd07v-ddtext.
  ENDIF.

ENDFORM.                    " GET_TYPECD_TEXTS

*&---------------------------------------------------------------------*
*&      Form  GET_GAS_PLANT_TEXTS
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  ---->  p1        text
*  <----  p2        text
*----------------------------------------------------------------------*
FORM get_gas_plant_texts.

  DATA: l_vname_nm LIKE oiu_do_jv-vname_nm.

  IF NOT roiu_pr_dn-bukrs IS INITIAL.
***********************************************************************
*              Select master data using full key                      *
***********************************************************************
    IF t001-bukrs NE roiu_pr_dn-bukrs.
      SELECT SINGLE * FROM t001   WHERE bukrs = roiu_pr_dn-bukrs.
      roiu_pr_dn-butxt = t001-butxt.
    ENDIF.
  ELSE.
    CLEAR roiu_pr_dn-butxt.
  ENDIF.

  IF NOT roiu_pr_dn-vname IS INITIAL.
    SELECT SINGLE * FROM  oiu_do_jv
                    WHERE bukrs = roiu_pr_dn-bukrs
                    AND   vname = roiu_pr_dn-vname.
```

```
1160
1161      IF sy-subrc NE 0.
1162        CLEAR w_fcode.
1163        PERFORM set_return_message USING c_messagetype_e
1164                                         c_messageid_cm
1165                                         '095'
1166                                         space
1167                                         space
1168                                         space
1169                                         space
1170                                         space
1171                                         space
1172                                         space
1173                                         space
1174                                         space.
1175      ENDIF.
1176
1177      oiu_do_jv-vname_nm = oiu_do_jv-vname_nm.
1178
1179    ELSE.
1180      CLEAR oiu_do_jv-vname_nm.
1181    ENDIF.
1182
1183
1184    IF NOT roiu_pr_dn-ref_dn_no IS INITIAL.
1185  ***********************************************************************
1186  *              Select master data using full key                      *
1187  ***********************************************************************
1188      IF oiu_pr_dn-dn_de NE roiu_pr_dn-ref_dn_de.
1189        SELECT SINGLE dn_de FROM oiu_pr_dn INTO roiu_pr_dn-ref_dn_de
1190                            WHERE dn_no = roiu_pr_dn-dn_no.
1191      ENDIF.
1192    ELSE.
1193      CLEAR roiu_pr_dn-ref_dn_de.
1194    ENDIF.
1195
1196    IF NOT roiu_pr_dn-gplt_dn_no IS INITIAL.
1197  ***********************************************************************
1198  *              Select master data using full key                      *
1199  ***********************************************************************
1200      IF oiu_pr_dn-dn_de NE roiu_pr_dn-gplt_dn_de.
1201        SELECT SINGLE dn_de FROM oiu_pr_dn INTO roiu_pr_dn-gplt_dn_de
1202                            WHERE dn_no = roiu_pr_dn-gplt_dn_no.
1203      ENDIF.
1204    ENDIF.
1205  ENDFORM.                    " GET_GAS_PLANT_TEXTS
1206  *&---------------------------------------------------------------------*
1207  *&      Form  SAVE_SUCCESSFUL
1208  *&---------------------------------------------------------------------*
1209  *       text
1210  *----------------------------------------------------------------------*
1211  *  -->  p1        text
1212  *  <--  p2        text
1213  *----------------------------------------------------------------------*
1214  FORM save_successful.
1215    DATA: l_text LIKE sy-msgv1.
1216
1217    DATA: f_dnno    LIKE roiu_pr_dn-dn_no,
1218          f_dnde    LIKE roiu_pr_dn-dn_de,
1219          f_dnal    LIKE roiu_pr_dnvd-al_fl,
1220          f_dnrc(2) TYPE c,
```

```
1221          f_dnid(8) TYPE c     VALUE 'OIUGT_DN'.
1222
1223    f_dnrc = '01'.
1224    f_dnno = roiu_pr_dn-dn_no.
1225    f_dnde = roiu_pr_dn-dn_de.
1226
1227    EXPORT f_dnrc f_dnno f_dnde f_dnal TO MEMORY ID f_dnid.
1228
1229    CASE w_mode.
1230      WHEN c_update.
1231        l_text = text-chg.
1232      WHEN c_create.
1233        l_text = text-crt.
1234    ENDCASE.
1235
1236    PERFORM set_return_message USING c_messagetype_s
1237                                     c_messageid_cm
1238                                     '006'
1239                                     space
1240                                     space
1241                                     text-002
1242                                     roiu_pr_dn-dn_no
1243                                     space
1244                                     l_text
1245                                     space
1246                                     space
1247                                     space.
1248
1249  ENDFORM.                    " SAVE_SUCCESSFUL
1250  *&---------------------------------------------------------------------*
1251  *&      Form  TRANSLATE_TEXTS
1252  *&---------------------------------------------------------------------*
1253  *       text
1254  *----------------------------------------------------------------------*
1255  *  -->  p1        text
1256  *  <--  p2        text
1257  *----------------------------------------------------------------------*
1258  FORM translate_texts.
1259
1260    DATA:    descr   LIKE roiu_pr_dn-dn_de.
1261
1262    descr = roiu_pr_dn-dn_de.
1263    TRANSLATE roiu_pr_dn-dn_de TO UPPER CASE.              "#EC TRANSLANG
1264    roiu_pr_dn-dn_de = descr.
1265
1266  ENDFORM.                    " TRANSLATE_TEXTS
1267  *&---------------------------------------------------------------------*
1268  *&      Form  NUMBER_RANGE_CHECK_BAPI
1269  *&---------------------------------------------------------------------*
1270  *       text
1271  *----------------------------------------------------------------------*
1272  *  -->  p1        text
1273  *  <--  p2        text
1274  *----------------------------------------------------------------------*
1275  FORM number_range_check_bapi.
1276
1277  *   Check to see if the number range group exists
1278  *    PERFORM check_for_number.
1279  *   Get the number range information in order to get the number range
1280    PERFORM get_number_range_info.
1281  *   If the number range is external lock the delivery network number
```

```
  IF w_interval_ds-externind = c_flag_on.
    PERFORM lock_delivery_number.
  ENDIF.
*   Checks to make sure if an internal number is selected
*   the delvery network field is blank
  PERFORM check_for_validity.
*   If the indicator is not external set del network number = INTER-001
  IF w_interval_ds-externind NE c_true.
    roiu_pr_dn-dn_no = c_internal.
  ENDIF.

ENDFORM.                    " NUMBER_RANGE_CHECK_BAPI
*&---------------------------------------------------------------------*
*&      Form  BAPI_EDIT_CHECKS
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  -->  p1        text
*  <--  p2        text
*----------------------------------------------------------------------*
FORM bapi_edit_checks.
  PERFORM check_dn_setid USING roiu_pr_dn-dn_set.
  PERFORM check_setid USING roiu_pr_dn-dn_set.
* Check for a duplicate delivery network
  PERFORM check_for_duplicate.
* Check for a duplicate DN in Razorfish OIUH_PR_DN table    "N 1732281
  PERFORM check_razorfish_dn_duplicate.                     "N 1732281
* Check that required field description is populated
  PERFORM check_description.
  PERFORM verify_entity_name.
* Check that required field DN type is populated and with a valid value
  PERFORM check_dn_type.
* Check that required field all. tol. is populated and with a valid valu
  PERFORM check_allocation_tolerance.
* Check for valid country and city information
  PERFORM check_country_and_region.
* Check that company code is populated when required and the value is
* valid
  PERFORM check_company_code.
* Check that joint venture is populated when required and the populated
* value exists within the correct company code
  PERFORM check_joint_venture.
* When the gas plant delivery network is filled make sure it is valid
  IF NOT roiu_pr_dn-gplt_dn_no IS INITIAL.
    PERFORM check_valid_gasplant_ntwk_id.
  ENDIF.

ENDFORM.                    " BAPI_EDIT_CHECKS

*&---------------------------------------------------------------------*
*&      Form  CHECK_DESCRIPTION
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  -->  p1        text
*  <--  p2        text
*----------------------------------------------------------------------*
FORM check_description.

  IF roiu_pr_dn-dn_de IS INITIAL.
    PERFORM set_return_message USING c_messagetype_e
```

```
                                        c_messageid_cm
                                        '008'
                                        space
                                        space
                                        text-025
                                        space
                                        space
                                        space
                                        space
                                        space
                                        space.
  ENDIF.

ENDFORM.                        " CHECK_DESCRIPTION

*&———————————————————————————————————————————————————————————————*
*&      Form  CHECK_ALLOCATION_TOLERANCE
*&———————————————————————————————————————————————————————————————*
*       text
*———————————————————————————————————————————————————————————————*
*  —>  p1        text
*  <—  p2        text
*———————————————————————————————————————————————————————————————*
FORM check_allocation_tolerance.

  IF roiu_pr_dn-tol_pc IS INITIAL.
    PERFORM set_return_message USING c_messagetype_e
                                     c_messageid_cm
                                     '008'
                                     space
                                     space
                                     text-026
                                     space
                                     space
                                     space
                                     space
                                     space
                                     space.
* Check to see if the allocation tolerance falls within the valid range
  ELSEIF ( roiu_pr_dn-tol_pc < 0 OR roiu_pr_dn-tol_pc > 999 ).
    PERFORM set_return_message USING c_messagetype_e
                                     c_messageid_cm
                                     '065'
                                     space
                                     space
                                     text-026
                                     c_zero
                                     c_lt_thousand
                                     space
                                     space
                                     space
                                     space.
  ENDIF.

ENDFORM.                        " CHECK_ALLOCATION_TOLERANCE

*&———————————————————————————————————————————————————————————————*
*&      Form  CHECK_DN_TYPE
*&———————————————————————————————————————————————————————————————*
*       text
*———————————————————————————————————————————————————————————————*
```

```
*  ——>  p1        text
*  <——  p2        text
*———————————————————————————————————————————————————————————————————*
FORM check_dn_type.

  IF roiu_pr_dn-type_cd IS INITIAL.
    PERFORM set_return_message USING c_messagetype_e
                                     c_messageid_cm
                                     '008'
                                     space
                                     space
                                     text-027
                                     space
                                     space
                                     space
                                     space
                                     space
                                     space.
  ELSEIF roiu_pr_dn-type_cd NE c_gas_plant
    AND  roiu_pr_dn-type_cd NE c_fuel_system
    AND  roiu_pr_dn-type_cd NE c_cluster
    AND  roiu_pr_dn-type_cd NE c_gathering_sys.
    PERFORM set_return_message USING c_messagetype_e
                                     c_messageid_cm
                                     '004'
                                     space
                                     space
                                     roiu_pr_dn-type_cd
                                     space
                                     text-027
                                     space
                                     space
                                     space
                                     space.
  ENDIF.

ENDFORM.                    " CHECK_DN_TYPE
*&———————————————————————————————————————————————————————————————————*
*&      Form  CHECK_COMPANY_CODE
*&———————————————————————————————————————————————————————————————————*
*       text
*———————————————————————————————————————————————————————————————————*
*  ——>  p1        text
*  <——  p2        text
*———————————————————————————————————————————————————————————————————*
FORM check_company_code.

  DATA:  l_type(132) TYPE c.

  IF roiu_pr_dn-type_cd = c_gas_plant
    AND NOT roiu_pr_dn-bukrs IS INITIAL.
*   Check to see if the company code is valid.
***********************************************************************
*              Select master data using full key                      *
***********************************************************************
    SELECT SINGLE * FROM t001 WHERE bukrs = roiu_pr_dn-bukrs.
    IF sy-subrc NE 0.
      PERFORM set_return_message USING c_messagetype_e
                                       c_messageid_cm
                                       '005'
                                       space
```

```
                                          space
                                          text-014
                                          roiu_pr_dn-bukrs
                                          space
                                          text-fnd
                                          space
                                          space
                                          space.
      ENDIF.
    ELSEIF roiu_pr_dn-type_cd = c_gas_plant
      AND  roiu_pr_dn-bukrs IS INITIAL.
      PERFORM set_return_message USING c_messagetype_e
                                       c_messageid_cm
                                       '008'
                                       space
                                       space
                                       text-012
                                       text-002
                                       space
                                       space
                                       space
                                       space
                                       space.
    ELSEIF roiu_pr_dn-type_cd NE c_gas_plant
      AND NOT roiu_pr_dn-bukrs IS INITIAL.
      CASE roiu_pr_dn-type_cd.
        WHEN c_gathering_sys.
          l_type = text-033.
        WHEN c_fuel_system.
          l_type = text-032.
        WHEN c_cluster.
          l_type = text-034.
      ENDCASE.
      PERFORM set_return_message USING c_messagetype_e
                                       c_messageid_cm
                                       '004'
                                       space
                                       space
                                       text-014
                                       space
                                       l_type
                                       text-002
                                       space
                                       space
                                       space.
    ENDIF.

  ENDFORM.                    " CHECK_COMPANY_CODE

  *&———————————————————————————————————————————————————————————————*
  *&      Form  CHECK_JOINT_VENTURE
  *&———————————————————————————————————————————————————————————————*
  *       text
  *———————————————————————————————————————————————————————————————*
  *  —>  p1        text
  *  <—  p2        text
  *———————————————————————————————————————————————————————————————*
  FORM check_joint_venture.

    DATA:  l_type(132)  TYPE c.
    TABLES: oiu_do_di.
```

```
  IF roiu_pr_dn-type_cd = c_gas_plant
    AND NOT roiu_pr_dn-bukrs IS INITIAL
    AND NOT roiu_pr_dn-vname IS INITIAL.
*********************************************************************
*              Select master data using full key                    *
*********************************************************************
    SELECT * FROM oiu_do_jv
    WHERE bukrs = roiu_pr_dn-bukrs
      AND vname = roiu_pr_dn-vname.
    ENDSELECT.


    IF sy-subrc NE 0.
      CLEAR w_fcode.
      PERFORM set_return_message USING c_messagetype_e
                                       c_messageid_cm
                                       '004'
                                       space
                                       space
                                       text-028
                                       roiu_pr_dn-vname
                                       text-014
                                       roiu_pr_dn-bukrs
                                       space
                                       space
                                       space.
    ELSEIF oiu_do_jv-type_cd <> '04'.
      CLEAR w_fcode.
      PERFORM set_return_message USING c_messagetype_e
                                      'OIUPR'
                                      '283'
                                      space
                                      space
                                      space
                                      space
                                      space
                                      space
                                      space
                                      space
                                      space.

    ENDIF.

  ELSEIF roiu_pr_dn-type_cd EQ c_gas_plant AND
     NOT roiu_pr_dn-bukrs IS INITIAL AND
         roiu_pr_dn-vname IS INITIAL.
    CLEAR w_fcode.
    PERFORM set_return_message USING c_messagetype_e
                                     c_messageid_cm
                                     '008'
                                     space
                                     space
                                     text-028
                                     space
                                     space
                                     space
                                     space
                                     space
                                     space.
  ELSEIF roiu_pr_dn-type_cd NE c_gas_plant
    AND NOT roiu_pr_dn-bukrs IS INITIAL
```

```
      AND NOT roiu_pr_dn-vname IS INITIAL.
      CASE roiu_pr_dn-type_cd.
        WHEN c_gathering_sys.
          l_type = text-033.
        WHEN c_fuel_system.
          l_type = text-032.
        WHEN c_cluster.
          l_type = text-034.
      ENDCASE.
      CLEAR w_fcode.
      PERFORM set_return_message USING c_messagetype_e
                                       c_messageid_cm
                                       '004'
                                       space
                                       space
                                       text-030
                                       space
                                       space
                                       l_type
                                       space
                                       space
                                       space.
    ENDIF.

  ENDFORM.                    " CHECK_JOINT_VENTURE
  *&---------------------------------------------------------------------*
  *&      Form  BAPI_EDIT_CHECKS_CHANGE
  *&---------------------------------------------------------------------*
  *       text
  *----------------------------------------------------------------------*
  *  -->  p1        text
  *  <--  p2        text
  *----------------------------------------------------------------------*
  FORM bapi_edit_checks_change.

  * Check that required field description is populated
    PERFORM check_description.
  * Check that required field all. tol. is populated and with a valid valu
    PERFORM check_allocation_tolerance.
  * Check for valid country and city information
    PERFORM check_country_and_region.
  * Check that company code is populated when required and the value is
  * valid
    PERFORM check_company_code.
  * Check that joint venture is populated when required and the populated
  * value exists within the correct company code
    PERFORM check_joint_venture.
  * When the gas plant delivery network is filled make sure it is valid
    IF NOT roiu_pr_dn-gplt_dn_no IS INITIAL.
      PERFORM check_valid_gasplant_ntwk_id.
    ENDIF.
  * Validate set id passed is a member of the standard hierarchy
    PERFORM check_dn_setid USING roiu_pr_dn-dn_set.
    PERFORM check_setid USING roiu_pr_dn-dn_set.

  ENDFORM.                    " BAPI_EDIT_CHECKS_CHANGE
  *&---------------------------------------------------------------------*
  *&      Form  SET_PARAMETER_ID
  *&---------------------------------------------------------------------*
  *       text
  *----------------------------------------------------------------------*
```

```
*   ——>  p1        text
*   <——  p2        text
*———————————————————————————————————————————————————————————————————*
FORM set_parameter_id.

  SET PARAMETER ID 'OIU_VENTURE_TYPE' FIELD c_gas_plant_jv.
  IF w_mode NE c_create.
    GET PARAMETER ID 'OIU_DN' FIELD roiu_pr_dn-dn_no.
    IF sy-subrc = 0.
      SELECT SINGLE dn_de FROM oiu_pr_dn INTO oiu_pr_dn-dn_de
                          WHERE dn_no = roiu_pr_dn-dn_no.
    ENDIF.
  ELSE.
    CLEAR roiu_pr_dn-dn_de.
    CLEAR setheadert-descript.           "SO7K011423 BT

    GET PARAMETER ID 'OIU_DN_GRP' FIELD roiu_pr_dn-dn_grp.
    IF sy-subrc = 0 AND
        sy-dynnr = c_screen_1100.
* single select using key fields
      SELECT SINGLE * FROM oiu_pr_dn_grp_tx
         WHERE dn_grp = roiu_pr_dn-dn_grp
          AND  spras  = sy-langu.
    ENDIF.
*check count of available groups, if one - set the default
    PERFORM check_single_grp.

    GET PARAMETER ID 'OIU_DN_TYPE_CD' FIELD roiu_pr_dn-type_cd.
    IF sy-subrc = 0.
      PERFORM get_typecd_texts.
    ENDIF.
    IF roiu_pr_dn-type_cd IS INITIAL AND
       w_mode = c_create.
      SET PARAMETER ID 'OIU_DN_TYPE_CD' FIELD c_type_cd.
    ENDIF.
    GET PARAMETER ID 'OIU_DN_REF' FIELD roiu_pr_dn-ref_dn_no.
    IF sy-subrc = 0.
      SELECT SINGLE dn_de FROM oiu_pr_dn INTO roiu_pr_dn-ref_dn_de
                          WHERE dn_no = roiu_pr_dn-ref_dn_no.
    ENDIF.
  ENDIF.

ENDFORM.                    " SET_PARAMETER_ID
*&———————————————————————————————————————————————————————————————————*
*&      Form  ROLLBACK_LONGTEXT
*&———————————————————————————————————————————————————————————————————*
*       text
*———————————————————————————————————————————————————————————————————*
*   ——>  p1        text
*   <——  p2        text
*———————————————————————————————————————————————————————————————————*
FORM rollback_longtext.

  IF NOT w_long_text_written_fg IS INITIAL.

    CALL FUNCTION 'DELETE_TEXT_FROM_CATALOG'
         EXPORTING
              object    = '*'
              name      = '*'
              id        = '*'
              language  = '*'
```

```
*        EXCEPTIONS
*             NOT_FOUND = 1
*             OTHERS    = 2
              .
  ENDIF.

ENDFORM.                    " ROLLBACK_LONGTEXT
*&---------------------------------------------------------------------*
*&      Form  CREATE_DELIVERY_NETWORK
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  -->  p1        text
*  <--  p2        text
*----------------------------------------------------------------------*
FORM create_delivery_network.


*  Before allowing save - check enqueue of standard hierarchy set
  PERFORM enqueue_setid USING space roiu_pr_dn-dn_set.
*  Get unique delivery network number for an internal number
*  assignment
  PERFORM get_del_network_number.
*  Check for Razorfish entry for internal Number range    "N 1736401
  PERFORM check_razorfish_dn_duplicate .                  "N 1736401
*  Lock the assigned delivery network number
  PERFORM lock_delivery_number.
*  Fill remaining internal table info for the save
  PERFORM prepare_create_for_save.
*  Call the function module to insert new record
  PERFORM save_del_net_data.
*  Commits data to the DB and Rollsback any bad insert
  PERFORM commit_all_work.
*  User message stating that the save was successful
  PERFORM save_successful.
* hierarchy update called separately
    IF SY-SUBRC = 0.                                     "SOGK027671 VV
     Perform F_UPDATE_HIERARCHY.                         "SOGK027671 VV
     COMMIT WORK.                                        "SOGK027671 VV
    ENDIF.                                               "SOGK027671 VV

  IF sy-calld = c_true OR sy-binpt = c_true.
    LEAVE TO SCREEN 0.
  ELSE.
    LEAVE TO TRANSACTION sy-tcode.
  ENDIF.

ENDFORM.                    " CREATE_DELIVERY_NETWORK
*&---------------------------------------------------------------------*
*&      Form  UPDATE_DELIVERY_NETWORK
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  -->  p1        text
*  <--  p2        text
*----------------------------------------------------------------------*
FORM update_delivery_network.

*  Before allowing save - check enqueue of DN Standard Hierarchy
  IF delivery_network_tb1-dn_set <> roiu_pr_dn-dn_set.
    PERFORM enqueue_setid USING delivery_network_tb1-dn_set
```

```
                                roiu_pr_dn-dn_set.
  ENDIF.
*  Fill remaining internal table info for the save
  PERFORM prepare_update_for_save.
*  Call the function module to insert new record
  PERFORM save_del_net_data.
*  Calls the change document function module using the old record
*  table and the changed record table
  PERFORM change_document USING delivery_network_tb1
                                roiu_pr_dn.
*  Commits all work to the DB and rollsback if bad update
  PERFORM commit_all_work.
  PERFORM save_successful.

* hierarchy update called separately
    IF SY-SUBRC = 0.                                        "SOGK027671 VV
     Perform F_UPDATE_HIERARCHY.                            "SOGK027671 VV
     COMMIT WORK.                                           "SOGK027671 VV
    ENDIF.                                                  "SOGK027671 VV



  IF sy-calld = c_true OR sy-binpt = c_true.
    LEAVE TO SCREEN 0.
  ELSE.
    LEAVE TO TRANSACTION sy-tcode.
  ENDIF.

ENDFORM.                    " UPDATE_DELIVERY_NETWORK
*&---------------------------------------------------------------------*
*&      Form  CHECK_INTITIAL_FIELDS
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  -->  p1        text
*  <--  p2        text
*----------------------------------------------------------------------*
*FORM check_intitial_fields.
*
*  IF roiu_pr_dn-tol_pc IS INITIAL.
*    PERFORM set_return_message USING c_messagetype_w
*                                     c_messageid_cm
*                                     '005'
*                                     space
*                                     space
*                                     text-026
*                                     space
*                                     space
*                                     text-029
*                                     space
*                                     space
*                                     space.
*  ENDIF.
*
*ENDFORM.                    " CHECK_INTITIAL_FIELDS
*&---------------------------------------------------------------------*
*&      Form  CLEAR_INFORMATION
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  -->  p1        text
```

```
*  <——  p2        text
*———————————————————————————————————————————————————————————————————*
FORM clear_information.

  CLEAR: roiu_pr_dn.

ENDFORM.                    " CLEAR_INFORMATION
*&———————————————————————————————————————————————————————————————————*
*&      Form  CHANGE_TO_DN_DATED
*&———————————————————————————————————————————————————————————————————*
*       text
*———————————————————————————————————————————————————————————————————*
*  ——>  p1        text
*  <——  p2        text
*———————————————————————————————————————————————————————————————————*
FORM change_to_dn_dated.

  CLEAR w_answer.
  IF w_mode NE c_display.
    IF delivery_network_tb1 NE roiu_pr_dn OR w_mode = c_create
       OR w_long_text_written_fg NE space.
      PERFORM popup_confirm_step USING text-003 text-004 text-006.
    ENDIF.
  ENDIF.
  IF w_answer <> c_cancel.
    PERFORM leave_to_dn_dated_trans.
  ENDIF.
ENDFORM.                    " CHANGE_TO_DN_DATED

*&———————————————————————————————————————————————————————————————————*
*&      Form  LEAVE_TO_DN_DATED_TRANS
*&———————————————————————————————————————————————————————————————————*
*       text
*———————————————————————————————————————————————————————————————————*
*  ——>  p1        text
*  <——  p2        text
*———————————————————————————————————————————————————————————————————*
FORM leave_to_dn_dated_trans.

  CASE w_fcode.
    WHEN c_fcode_dnd1.
      LEAVE TO TRANSACTION c_trans_dnd1.
    WHEN c_fcode_dnd2.
      LEAVE TO TRANSACTION c_trans_dnd2.
    WHEN c_fcode_dnd3.
      LEAVE TO TRANSACTION c_trans_dnd3.
    WHEN c_fcode_dnd6.
      LEAVE TO TRANSACTION c_trans_dnd6.
  ENDCASE.

ENDFORM.                    " LEAVE_TO_DN_DATED_TRANS
*&———————————————————————————————————————————————————————————————————*
*&      Form  CHECK_COMBOBOX_STATUS
*&———————————————————————————————————————————————————————————————————*
*  CHECKING THE STATUS OF THE COMBOBOXES ON THE SCREEN TO SEE IF THEY
*  HAVE CHANGED BASED ON THE FUNCTION CODE.  IF THE COMBOBOXES HAVE
*  CHANGED, THEN DEPENDENT COMBOBOXES NEED TO BE CLEARED AND
*  RE-POPULATED
*———————————————————————————————————————————————————————————————————*
FORM check_combobox_status.

```

```
  IF w_cntry <> roiu_pr_dn-cntry_cd.
    CLEAR roiu_pr_dn-pri_geo_loc.
    CLEAR roiu_pr_dn-sec_geo_loc.
    w_fcode = c_fcode_loc.
  ELSEIF w_primary <> roiu_pr_dn-pri_geo_loc.
    CLEAR roiu_pr_dn-sec_geo_loc.
    w_fcode = c_fcode_loc.
  ENDIF.

ENDFORM.                    " CHECK_COMBOBOX_STATUS

*&———————————————————————————————————————————————————————————————*
*&      Form  CHECK_SINGLE_GRP
*&———————————————————————————————————————————————————————————————*
*       text
*———————————————————————————————————————————————————————————————*
*  —>  p1        text
*  <—  p2        text
*———————————————————————————————————————————————————————————————*
FORM check_single_grp.
*This sets the group default if only one group exists
  DATA: BEGIN OF grp_tb OCCURS 0.
          INCLUDE STRUCTURE oiu_pr_dn_grp.
  DATA: END OF grp_tb.

  SELECT * FROM oiu_pr_dn_grp
    APPENDING TABLE grp_tb.

  IF sy-subrc = 0.
    DESCRIBE TABLE grp_tb LINES sy-tfill.
    IF sy-tfill = 1.
      READ TABLE grp_tb.
      IF g_bapi_fg EQ c_true.
        roiu_pr_dn-dn_grp = grp_tb-dn_grp.
      ELSE.
        SET PARAMETER ID 'OIU_DN_GRP' FIELD grp_tb-dn_grp.
      ENDIF.
    ENDIF.
  ENDIF.

ENDFORM.                    " CHECK_SINGLE_GRP

*&———————————————————————————————————————————————————————————————*
*&      Form  CHECK_SETID
*&———————————————————————————————————————————————————————————————*
* VALIDATE THE SET NAME THAT IS ENTERED BY THE USER.
*   1.  CHECK THAT SET EXISTS.
*   2.  CHECK THAT THE SET IS A MEMBER OF THE STANDARD HIERARCHY
*———————————————————————————————————————————————————————————————*
*      —>P_SET    SET NAME ENTERED BY THE USER ON SCREEN
*———————————————————————————————————————————————————————————————*
FORM check_setid USING    set LIKE roiu_pr_dn-dn_set.

  DATA: l_setid LIKE sethier-setid.
  DATA: l_setname TYPE setnamenew.
  DATA: ls_setinfo LIKE setinfo.


  CLEAR setheadert-descript.                "SO7K011423 BT
  l_setname = set.

```

```
  CALL FUNCTION 'G_SET_ENCRYPT_SETID'
       EXPORTING
            setclass             = c_setclass
            shortname            = l_setname
*           KOKRS                =
*           KTOPL                =
*           LIB                  =
*           RNAME                =
*           ECCS_DIMEN           =
*           ECCS_ITCLG           =
*           ECCS_SITYP           =
       IMPORTING
            setid                = l_setid.
*       EXCEPTIONS
*            NO_CO_AREA_SPECIFIED = 1
*            ILLEGAL_SETCLASS     = 2
*            OTHERS               = 3
*            .
*

* —— check the type
  CALL FUNCTION 'G_SET_GET_INFO'
       EXPORTING
            setname = l_setid
       IMPORTING
            info    = ls_setinfo
       EXCEPTIONS
            OTHERS  = 1.

  IF sy-subrc <> 0.
    PERFORM set_return_message
                USING
                   c_messagetype_e
                   c_messageid_cm
                   '200'
                   space
                   space
                   set
                   space
                   space
                   space
                   space
                   space
                   space.

  ENDIF.

  setheadert-descript = ls_setinfo-title.   " SO7K011423 BT

  IF ls_setinfo-type NA 'BS'.
    PERFORM set_return_message
            USING
               c_messagetype_e
               c_messageid_cm
               '201'
               space
               space
               set
               space
               space
               space
```

```
                 space
                 space
                 space.

    ELSE.
      CALL FUNCTION 'OIU_CHECK_HIER_IN_STD'
           EXPORTING
                i_setclass      = c_setclass
                i_setname       = l_setname
           EXCEPTIONS
                not_in_standard = 1
                OTHERS          = 2.
      IF sy-subrc <> 0.
        PERFORM set_return_message
                  USING
                     c_messagetype_e
                     c_messageid_cm
                     '202'
                     space
                     space
                     set
                     space
                     space
                     space
                     space
                     space
                     space.
      ENDIF.
    ENDIF.

*  setheadert-descript = ls_setinfo-title.     SO7K011423 BT

ENDFORM.                    " CHECK_SETID

*&———————————————————————————————————————————————————————————————*
*&      Form  CHECK_DN_SETID
*&———————————————————————————————————————————————————————————————*
*       text
*———————————————————————————————————————————————————————————————*
*      ——>P_ROIU_PR_DN_DN_SET  text
*———————————————————————————————————————————————————————————————*
FORM check_dn_setid USING    set LIKE roiu_pr_dn-dn_set.
  IF set IS INITIAL.
    PERFORM set_return_message USING c_messagetype_e
                                     c_messageid_cm
                                     '008'
                                     space
                                     space
                                     text-hie
                                     space
                                     space
                                     space
                                     space
                                     space
                                     space.
  ENDIF.

ENDFORM.                    " CHECK_DN_SETID

*&———————————————————————————————————————————————————————————————*
*&      Form  GET_SETID_DESC
```

```
*&---------------------------------------------------------------------*
*   1.  CHECK THAT SET EXISTS.
*   2.  CHECK THAT THE SET IS A MEMBER OF THE STANDARD HIERARCHY
*   3.  Get the SETID's description
*---------------------------------------------------------------------*
FORM get_setid_desc.

  DATA: l_setid    LIKE sethier-setid.
  DATA: l_setname  TYPE setnamenew.
  DATA: ls_setinfo LIKE setinfo.

  CLEAR setheadert-descript.

  l_setname = roiu_pr_dn-dn_set.

  CALL FUNCTION 'G_SET_ENCRYPT_SETID'
       EXPORTING
            setclass  = c_setclass
            shortname = l_setname
       IMPORTING
            setid     = l_setid.

* -- check the type
  CALL FUNCTION 'G_SET_GET_INFO'
       EXPORTING
            setname = l_setid
       IMPORTING
            info    = ls_setinfo
       EXCEPTIONS
            OTHERS  = 1.

  IF sy-subrc <> 0.
    EXIT.
  ENDIF.

  setheadert-descript = ls_setinfo-title.

ENDFORM.                    " GET_SETID_DESC
```

```
*eject
*&---------------------------------------------------------------------*
*&      Form  ENQUEUE_SETID
*&---------------------------------------------------------------------*
*       Lock Set ID's for Update - OLD and NEW prior to allowing save
*----------------------------------------------------------------------*
*  ---->  OLD_SET   Old Set ID from Record (Create = Blank)
*  ---->  NEW_SET   New Set ID from Record
*----------------------------------------------------------------------*
FORM enqueue_setid USING p_old
                         p_new.

  DATA: l_setname TYPE setnamenew.
  DATA: l_setid   TYPE setid.

* is the Hier. active?
  m_hier_active.                                          " SOGK013053 EY
* If "NO" then skip the hierarchy update.
  IF w_hier_active IS INITIAL.                            " SOGK013053 EY
    EXIT.                                                " SOGK013053 EY
  ENDIF.                                                  " SOGK013053 EY

  IF NOT ( p_old IS INITIAL ).

    l_setname = p_old.

    CALL FUNCTION 'G_SET_ENCRYPT_SETID'
         EXPORTING
              setclass             = c_setclass
              shortname            = l_setname
*               KOKRS                =
*               KTOPL                =
*               LIB                  =
*               RNAME                =
*               ECCS_DIMEN           =
*               ECCS_ITCLG           =
*               ECCS_SITYP           =
         IMPORTING
              setid                = l_setid.
*           EXCEPTIONS
*                NO_CO_AREA_SPECIFIED = 1
*                ILLEGAL_SETCLASS     = 2
*                OTHERS               = 3


    CALL FUNCTION 'G_SET_ENQUEUE_WITH_FATHERS'
         EXPORTING
              setid          = l_setid
         EXCEPTIONS
              foreign_lock   = 1
              system_failure = 2
              OTHERS         = 3.

    IF sy-subrc <> 0.
      CLEAR w_fcode.
      PERFORM set_return_message
                  USING
                     c_messagetype_e
                     c_messageid_cm
                     '203'
                     space
```

```
                      space
                      space
                      space
                      space
                      space
                      space
                      space
                      space.
      ENDIF.

    ENDIF.

    IF NOT ( p_new IS INITIAL ).

      l_setname = p_new.

      CALL FUNCTION 'G_SET_ENCRYPT_SETID'
           EXPORTING
                setclass             = c_setclass
                shortname            = l_setname
*               KOKRS                =
*               KTOPL                =
*               LIB                  =
*               RNAME                =
*               ECCS_DIMEN           =
*               ECCS_ITCLG           =
*               ECCS_SITYP           =
           IMPORTING
                setid                = l_setid.
*             EXCEPTIONS
*                  NO_CO_AREA_SPECIFIED = 1
*                  ILLEGAL_SETCLASS     = 2
*                  OTHERS               = 3


      CALL FUNCTION 'G_SET_ENQUEUE_WITH_FATHERS'
           EXPORTING
                setid          = l_setid
           EXCEPTIONS
                foreign_lock   = 1
                system_failure = 2
                OTHERS         = 3.

      IF sy-subrc <> 0.
        CLEAR w_fcode.
        PERFORM set_return_message
                    USING
                       c_messagetype_e
                       c_messageid_cm
                       '203'
                       space
                       space
                       space
                       space
                       space
                       space
                       space
                       space
                       space.
      ENDIF.

```

```
  ENDIF.



ENDFORM.                    " ENQUEUE_SETID
*&---------------------------------------------------------------------*
*&      Form  GET_DN_GRP_TX
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*      -->P_ROIU_PR_DN_DN_GRP  text
*      -->P_L_DN_GRP_TX  text
*----------------------------------------------------------------------*
FORM get_dn_grp_tx USING    p_dn_grp
                            p_dn_grp_tx.

  IF NOT p_dn_grp IS INITIAL.
    SELECT SINGLE dn_grp_tx
    INTO   p_dn_grp_tx
    FROM   oiu_pr_dn_grp_tx
    WHERE  spras  = sy-langu  AND
           dn_grp = p_dn_grp.

    IF sy-subrc NE 0.
      CLEAR p_dn_grp_tx.
    ENDIF.
  ELSE.
    CLEAR p_dn_grp_tx.
  ENDIF.

ENDFORM.                    " GET_DN_GRP_TX

*&---------------------------------------------------------------------*
*&      Form  DEFAULT_HIER_NODE
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
*  -->  p1        text
*  <--  p2        text
*----------------------------------------------------------------------*
FORM default_hier_node.
  DATA: l_single_dn_set  TYPE setidlist,
        l_setid          LIKE sethier-setid,
        l_line_count     LIKE sy-index.             "SOGK005579 BT

  CALL FUNCTION 'OIU_SET_STD_HIER_NODES'
       EXPORTING
            setclass     = c_setclass
       IMPORTING
            single_set   = l_single_dn_set
            line_count   = l_line_count  "SOGK005579 BT
       EXCEPTIONS
            default_node = 1
            OTHERS       = 2.

  CASE sy-subrc.
    WHEN 1.
      READ TABLE l_single_dn_set INTO l_setid INDEX 1.
      roiu_pr_dn-dn_set = l_setid+4.
      PERFORM check_setid USING roiu_pr_dn-dn_set.  "SO7K011423 BT
      IF l_line_count > 1.                          "SOGK005579 BT
```

```
2297          CLEAR roiu_pr_dn-dn_set.                    "SOGK005579 BT
2298        ENDIF.                                        "SOGK005579 BT
2299      WHEN OTHERS.
2300        CLEAR roiu_pr_dn-dn_set.
2301    ENDCASE.
2302
2303  ENDFORM.                    " DEFAULT_HIER_NODE
2304
2305  *&———————————————————————————————————————————————————————————————*
2306  *&      Form  VERIFY_ENTITY_NAME
2307  *&———————————————————————————————————————————————————————————————*
2308  *       text
2309  *———————————————————————————————————————————————————————————————*
2310  *  —>  p1        text
2311  *  <—  p2        text
2312  *———————————————————————————————————————————————————————————————*
2313  FORM verify_entity_name.
2314    DATA: l_oiu_cm_edit LIKE oiu_cm_edit.
2315  *KH  SELECT * FROM oiu_cm_edit
2316    SELECT SINGLE * FROM oiu_cm_edit              "KH
2317    INTO l_oiu_cm_edit
2318        WHERE edit_no = '0009'.
2319  *KH  ENDSELECT.
2320    CHECK l_oiu_cm_edit-status = c_true.
2321    SET LOCALE LANGUAGE sy-langu.
2322    roiu_pr_dn-shtxt = roiu_pr_dn-dn_de.
2323    TRANSLATE roiu_pr_dn-shtxt TO UPPER CASE.              "#EC TRANSLANG
2324    SET LOCALE LANGUAGE space.
2325    SELECT * FROM oiu_pr_dn
2326    WHERE shtxt = roiu_pr_dn-shtxt
2327    AND   dn_no <> roiu_pr_dn-dn_no.
2328    ENDSELECT.
2329    IF sy-subrc = 0.
2330      CLEAR w_fcode.
2331      PERFORM set_return_message USING 'E'
2332                                       'OIUCM'
2333                                       '100'
2334                                       space
2335                                       space
2336                                       text-cme
2337                                       text-dup
2338                                       oiu_pr_dn-dn_no
2339                                       space
2340                                       space
2341                                       space
2342                                       space.
2343
2344    ENDIF.
2345
2346  ENDFORM.                    " VERIFY_ENTITY_NAME
2347
2348  *&———————————————————————————————————————————————————————————————*
2349  *&      Form  F_UPDATE_HIERARCHY
2350  *&———————————————————————————————————————————————————————————————*
2351  *       text
2352  *———————————————————————————————————————————————————————————————*
2353  FORM f_update_hierarchy.
2354
2355  * SOGK013053  EY  3/13/02  OSS 126627/2002  Hierarchy performance review
2356
2357  * is the Hier. active?
```

```
  m_hier_active.                                          " SOGK013053 EY

  CHECK delivery_network_tb1-dn_set <> roiu_pr_dn-dn_set
     OR w_mode                       = c_create.

  IF w_mode    = c_create.
    w_old_set = space.
  ELSE.
    w_old_set = delivery_network_tb1-dn_set.
  ENDIF.

  w_new_set    = roiu_pr_dn-dn_set.
  w_set_object = roiu_pr_dn-dn_no.

* If "Not Active" then skip the hierarchy update.
  IF w_hier_active IS INITIAL.                            " SOGK013053 EY
  ELSE.                                                   " SOGK013053 EY
    CALL FUNCTION 'OIU_UPDATE_HIERARCHY' IN UPDATE TASK
         EXPORTING
              setclass = c_setclass
              old_set  = w_old_set
              new_set  = w_new_set
              object   = w_set_object.
  ENDIF.                                                  " SOGK013053 EY

  CALL FUNCTION 'OIU_UPDATE_CM_HIER' IN UPDATE TASK " SOGK013053 EY
       EXPORTING
            i_setcls = c_setclass
            i_oldset = w_old_set
            i_newset = w_new_set
            i_object = w_set_object.

ENDFORM.                    " F_UPDATE_HIERARCHY
*&---------------------------------------------------------------------*
*&      Form  CHECK_PREMAS_DN_DUPLICATE
*&---------------------------------------------------------------------*
* checks the existence of DN number in razorfish table OIUH_PR_DN
* Note # 1732281
*---------------------------------------------------------------------*
*  ---> p1        text
*  <--- p2        text
*---------------------------------------------------------------------*
form CHECK_RAZORFISH_DN_DUPLICATE .

  DATA : lv_dn_no LIKE OIUH_PR_DN-DN_DN_NO ,
         lv_sub_dn LIKE   OIUH_PR_DN-DN_SUB_NWK_NO .
  DATA : lt_oiuh_pr_dn TYPE TABLE OF OIUH_PR_DN .
  DATA : ls_oiuh_pr_dn LIKE LINE OF lt_oiuh_pr_dn .
* the last 4 char as sub dn number
  lv_sub_dn = roiu_pr_dn-dn_no+16(4).
* char 5 to 8 used as dn number
  lv_dn_no  = roiu_pr_dn-dn_no+12(4).

  IF w_mode = c_create AND NOT roiu_pr_dn-dn_no IS INITIAL.
     SELECT SINGLE *
        INTO ls_oiuh_pr_dn
        FROM OIUH_PR_DN
       WHERE DN_DN_NO      = lv_dn_no AND
             DN_SUB_NWK_NO = lv_sub_dn .
     IF sy-subrc = 0.
*      Start of "N 1736401
```

```
        IF w_interval_ds-externind IS INITIAL.
           PERFORM set_return_message USING c_messagetype_e
                                            c_messageid_cm
                                            '100'
                                            space
                                            space
                                            text-037
                                            space
                                            space
                                            space
                                            space
                                            space
                                            space.
        ELSE.
*       End of Note "N 1736401
           PERFORM set_return_message USING c_messagetype_e
                                            c_messageid_cm
                                          '003'
                                          space
                                          space
                                          text-036
                                          text-002
                                          roiu_pr_dn-dn_no
                                          space
                                          space
                                          space
                                          space.
       ENDIF.                                        "N 1736401
     ENDIF.
   ENDIF.
endform.                      " CHECK_PREMAS_DN_DUPLICATE
```

Subroutines:		Line	Page
ADMINISTRATIVE_DATA.....	LOIU_DELIVERY_NETWORKF01	577	34
BAPI_EDIT_CHECKS.....	LOIU_DELIVERY_NETWORKF01	1302	46
BAPI_EDIT_CHECKS_CHANGE.....	LOIU_DELIVERY_NETWORKF01	1620	51
CD_CALL_OIU_DN.....	LOIU_DELIVERY_NETWORKF01	238	28
CHANGE_DOCUMENT.....	LOIU_DELIVERY_NETWORKF01	225	28
	LOIU_DELIVERY_NETWORKF01	1778	54
CHANGE_TO_DN_DATED.....	LOIU_DELIVERY_NETWORKF01	1846	55
CHECK_ALLOCATION_TOLERANCE....	LOIU_DELIVERY_NETWORKF01	1315	46
	LOIU_DELIVERY_NETWORKF01	1366	47
	LOIU_DELIVERY_NETWORKF01	1625	51
CHECK_COMBOBOX_STATUS.....	LOIU_DELIVERY_NETWORKF01	1890	55
CHECK_COMPANY_CODE.....	LOIU_DELIVERY_NETWORKF01	1320	46
	LOIU_DELIVERY_NETWORKF01	1449	48
	LOIU_DELIVERY_NETWORKF01	1630	51
CHECK_COUNTRY_AND_REGION.....	LOIU_DELIVERY_NETWORKF01	742	37
	LOIU_DELIVERY_NETWORKF01	1317	46
	LOIU_DELIVERY_NETWORKF01	1627	51
CHECK_DESCRIPTION.....	LOIU_DELIVERY_NETWORKF01	1310	46
	LOIU_DELIVERY_NETWORKF01	1339	46
	LOIU_DELIVERY_NETWORKF01	1623	51
CHECK_DN_SETID.....	LOIU_DELIVERY_NETWORKF01	1303	46
	LOIU_DELIVERY_NETWORKF01	1639	51
	LOIU_DELIVERY_NETWORKF01	2055	58
CHECK_DN_TYPE.....	LOIU_DELIVERY_NETWORKF01	1313	46
	LOIU_DELIVERY_NETWORKF01	1407	48
CHECK_FOR_CHANGES.....	LOIU_DELIVERY_NETWORKF01	162	27
CHECK_FOR_DUPLICATE.....	LOIU_DELIVERY_NETWORKF01	533	33
	LOIU_DELIVERY_NETWORKF01	1306	46
CHECK_FOR_GAS_PLANT.....	LOIU_DELIVERY_NETWORKF01	606	34
CHECK_FOR_NUMBER.....	LOIU_DELIVERY_NETWORKF01	275	29
CHECK_FOR_VALIDITY.....	LOIU_DELIVERY_NETWORKF01	363	30
	LOIU_DELIVERY_NETWORKF01	1287	46
CHECK_JOINT_VENTURE.....	LOIU_DELIVERY_NETWORKF01	674	36
	LOIU_DELIVERY_NETWORKF01	1323	46
	LOIU_DELIVERY_NETWORKF01	1522	49
	LOIU_DELIVERY_NETWORKF01	1633	51
CHECK_RAZORFISH_DN_DUPLICATE..	LOIU_DELIVERY_NETWORKF01	1308	46
	LOIU_DELIVERY_NETWORKF01	1733	53
	LOIU_DELIVERY_NETWORKF01	2400	64
CHECK_SETID.....	LOIU_DELIVERY_NETWORKF01	1304	46
	LOIU_DELIVERY_NETWORKF01	1640	51
	LOIU_DELIVERY_NETWORKF01	1943	56
	LOIU_DELIVERY_NETWORKF01	2295	62
CHECK_SINGLE_GRP.....	LOIU_DELIVERY_NETWORKF01	1673	52
	LOIU_DELIVERY_NETWORKF01	1911	56
CHECK_VALID_GASPLANT_NTWK_ID..	LOIU_DELIVERY_NETWORKF01	684	36
	LOIU_DELIVERY_NETWORKF01	1326	46
	LOIU_DELIVERY_NETWORKF01	1636	51
CLEAR_INFORMATION.....	LOIU_DELIVERY_NETWORKF01	1833	55
COMMIT_ALL_WORK.....	LOIU_DELIVERY_NETWORKF01	1022	41
	LOIU_DELIVERY_NETWORKF01	1741	53
	LOIU_DELIVERY_NETWORKF01	1781	54
CREATE_DELIVERY_NETWORK.....	LOIU_DELIVERY_NETWORKF01	207	28
	LOIU_DELIVERY_NETWORKF01	1724	53
DEFAULT_HIER_NODE.....	LOIU_DELIVERY_NETWORKF01	2276	62
ENQUEUE_SETID.....	LOIU_DELIVERY_NETWORKF01	1728	53
	LOIU_DELIVERY_NETWORKF01	1769	53
	LOIU_DELIVERY_NETWORKF01	2123	60
F_UPDATE_HIERARCHY.....	LOIU_DELIVERY_NETWORKF01	1746	53

```
                                LOIU_DELIVERY_NETWORKF01      1786    54
                                LOIU_DELIVERY_NETWORKF01      2353    63
GET_COUNTRY_TEXTS............. LOIU_DELIVERY_NETWORKF01      1060    42
GET_DEL_NETWORK_NUMBER........ LOIU_DELIVERY_NETWORKF01       462    32
                                LOIU_DELIVERY_NETWORKF01      1731    53
GET_DN_GRP_TX................. LOIU_DELIVERY_NETWORKF01       372    31
                                LOIU_DELIVERY_NETWORKF01      2249    62
GET_GAS_PLANT_TEXTS........... LOIU_DELIVERY_NETWORKF01      1140    43
GET_NUMBER_RANGE_INFO......... LOIU_DELIVERY_NETWORKF01       289    29
                                LOIU_DELIVERY_NETWORKF01      1280    45
GET_REF_DELIVERY_NETWORK...... LOIU_DELIVERY_NETWORKF01        12    25
GET_SETID_DESC................ LOIU_DELIVERY_NETWORKF01        59    25
                                LOIU_DELIVERY_NETWORKF01      2080    59
GET_TYPECD_TEXTS.............. LOIU_DELIVERY_NETWORKF01      1109    43
                                LOIU_DELIVERY_NETWORKF01      1677    52
LEAVE_TO_DN_DATED_TRANS....... LOIU_DELIVERY_NETWORKF01      1856    55
                                LOIU_DELIVERY_NETWORKF01      1868    55
LEAVE_TO_TRANSACTION.......... LOIU_DELIVERY_NETWORKF01       903    39
LOCK_DELIVERY_NUMBER.......... LOIU_DELIVERY_NETWORKF01       816    38
                                LOIU_DELIVERY_NETWORKF01      1283    46
                                LOIU_DELIVERY_NETWORKF01      1735    53
NO_CHANGES_MADE............... LOIU_DELIVERY_NETWORKF01       250    29
NUMBER_RANGE_CHECK_BAPI....... LOIU_DELIVERY_NETWORKF01      1275    45
POPUP_CONFIRM_STEP............ LOIU_DELIVERY_NETWORKF01       171    27
                                LOIU_DELIVERY_NETWORKF01       173    27
                                LOIU_DELIVERY_NETWORKF01       188    28
                                LOIU_DELIVERY_NETWORKF01      1852    55
PREPARE_CREATE_FOR_SAVE....... LOIU_DELIVERY_NETWORKF01       120    26
                                LOIU_DELIVERY_NETWORKF01      1737    53
PREPARE_LONG_TEXT............. LOIU_DELIVERY_NETWORKF01      1036    41
PREPARE_UPDATE_FOR_SAVE....... LOIU_DELIVERY_NETWORKF01       144    27
                                LOIU_DELIVERY_NETWORKF01      1773    54
ROLLBACK_LONGTEXT............. LOIU_DELIVERY_NETWORKF01       212    28
                                LOIU_DELIVERY_NETWORKF01      1699    52
SAVE_DEL_NET_DATA............. LOIU_DELIVERY_NETWORKF01        89    26
                                LOIU_DELIVERY_NETWORKF01      1739    53
                                LOIU_DELIVERY_NETWORKF01      1775    54
SAVE_SUCCESSFUL............... LOIU_DELIVERY_NETWORKF01      1214    44
                                LOIU_DELIVERY_NETWORKF01      1743    53
                                LOIU_DELIVERY_NETWORKF01      1782    54
SET_PARAMETER_ID.............. LOIU_DELIVERY_NETWORKF01      1651    52
SET_RETURN_MESSAGE............ LOIU_DELIVERY_NETWORKF01        36    25
                                LOIU_DELIVERY_NETWORKF01       252    29
                                LOIU_DELIVERY_NETWORKF01       298    29
                                LOIU_DELIVERY_NETWORKF01       341    30
                                LOIU_DELIVERY_NETWORKF01       377    31
                                LOIU_DELIVERY_NETWORKF01       419    31
                                LOIU_DELIVERY_NETWORKF01       508    33
                                LOIU_DELIVERY_NETWORKF01       553    34
                                LOIU_DELIVERY_NETWORKF01       624    35
                                LOIU_DELIVERY_NETWORKF01       642    35
                                LOIU_DELIVERY_NETWORKF01       661    35
                                LOIU_DELIVERY_NETWORKF01       704    36
                                LOIU_DELIVERY_NETWORKF01       718    36
                                LOIU_DELIVERY_NETWORKF01       761    37
                                LOIU_DELIVERY_NETWORKF01       793    37
                                LOIU_DELIVERY_NETWORKF01       834    38
                                LOIU_DELIVERY_NETWORKF01       847    38
                                LOIU_DELIVERY_NETWORKF01       860    39
                                LOIU_DELIVERY_NETWORKF01       917    40
                                LOIU_DELIVERY_NETWORKF01      1163    44
```

	LOIU_DELIVERY_NETWORKF01	1236	45
	LOIU_DELIVERY_NETWORKF01	1342	46
	LOIU_DELIVERY_NETWORKF01	1369	47
	LOIU_DELIVERY_NETWORKF01	1383	47
	LOIU_DELIVERY_NETWORKF01	1410	48
	LOIU_DELIVERY_NETWORKF01	1426	48
	LOIU_DELIVERY_NETWORKF01	1461	48
	LOIU_DELIVERY_NETWORKF01	1476	49
	LOIU_DELIVERY_NETWORKF01	1498	49
	LOIU_DELIVERY_NETWORKF01	1540	50
	LOIU_DELIVERY_NETWORKF01	1554	50
	LOIU_DELIVERY_NETWORKF01	1573	50
	LOIU_DELIVERY_NETWORKF01	1597	51
	LOIU_DELIVERY_NETWORKF01	1983	57
	LOIU_DELIVERY_NETWORKF01	2003	57
	LOIU_DELIVERY_NETWORKF01	2027	58
	LOIU_DELIVERY_NETWORKF01	2057	58
	LOIU_DELIVERY_NETWORKF01	2169	60
	LOIU_DELIVERY_NETWORKF01	2220	61
	LOIU_DELIVERY_NETWORKF01	2331	63
	LOIU_DELIVERY_NETWORKF01	2420	65
	LOIU_DELIVERY_NETWORKF01	2434	65
TRANSLATE_TEXTS.....	LOIU_DELIVERY_NETWORKF01	1258	45
UNLOCK_DELIVERY_NUMBER.....	LOIU_DELIVERY_NETWORKF01	213	28
	LOIU_DELIVERY_NETWORKF01	404	31
	LOIU_DELIVERY_NETWORKF01	882	39
UPDATE_DELIVERY_NETWORK.....	LOIU_DELIVERY_NETWORKF01	209	28
	LOIU_DELIVERY_NETWORKF01	1765	53
USE_LONG_TEXT_PROCESSING.....	LOIU_DELIVERY_NETWORKF01	940	40
	LOIU_DELIVERY_NETWORKF01	1043	42
VERIFY_ENTITY_NAME.....	LOIU_DELIVERY_NETWORKF01	1311	46
	LOIU_DELIVERY_NETWORKF01	2313	63

Messages:

		Line	Page
036.....	LOIU_DELIVERY_NETWORKF01	981	41

External calls

Function modules:		Line	Page
COMMIT_TEXT.....	LOIU_DELIVERY_NETWORKF01	1024	41
DELETE_TEXT_FROM_CATALOG.....	LOIU_DELIVERY_NETWORKF01	1703	52
DEQUEUE_E_OIU_DN.....	LOIU_DELIVERY_NETWORKF01	884	39
DOMAIN_VALUE_GET.....	LOIU_DELIVERY_NETWORKF01	1116	43
ENQUEUE_E_OIU_DN.....	LOIU_DELIVERY_NETWORKF01	818	38
G_SET_ENCRYPT_SETID.....	LOIU_DELIVERY_NETWORKF01	1953	57
	LOIU_DELIVERY_NETWORKF01	2090	59
	LOIU_DELIVERY_NETWORKF01	2140	60
	LOIU_DELIVERY_NETWORKF01	2191	61
G_SET_ENQUEUE_WITH_FATHERS....	LOIU_DELIVERY_NETWORKF01	2159	60
	LOIU_DELIVERY_NETWORKF01	2210	61
G_SET_GET_INFO.....	LOIU_DELIVERY_NETWORKF01	1974	57
	LOIU_DELIVERY_NETWORKF01	2098	59
IDOC_OUTPUT_OIU_DELIVERY_NETWK	LOIU_DELIVERY_NETWORKF01	106	26
OIU_CHECK_HIER_IN_STD.....	LOIU_DELIVERY_NETWORKF01	2019	58
OIU_CM_CREATE_GROUP_INFO.....	LOIU_DELIVERY_NETWORKF01	594	34
OIU_DN_MASTER_READ.....	LOIU_DELIVERY_NETWORKF01	25	25
	LOIU_DELIVERY_NETWORKF01	542	33
	LOIU_DELIVERY_NETWORKF01	692	36
OIU_GET_NEXT_SYS_NO.....	LOIU_DELIVERY_NETWORKF01	492	33
OIU_LEAVE_TXN_CTL.....	LOIU_DELIVERY_NETWORKF01	906	39
OIU_LONG_TEXT_MAINTAIN.....	LOIU_DELIVERY_NETWORKF01	950	40
	LOIU_DELIVERY_NETWORKF01	969	40
	LOIU_DELIVERY_NETWORKF01	986	41
	LOIU_DELIVERY_NETWORKF01	999	41
OIU_NODE_DN_INSERT.....	LOIU_DELIVERY_NETWORKF01	97	26
OIU_NUMBER_GET_INFO.....	LOIU_DELIVERY_NETWORKF01	327	30
OIU_SET_STD_HIER_NODES.....	LOIU_DELIVERY_NETWORKF01	2281	62
OIU_UPDATE_CM_HIER.....	LOIU_DELIVERY_NETWORKF01	2383	64
OIU_UPDATE_DELIVERY_NETWORK...	LOIU_DELIVERY_NETWORKF01	91	26
OIU_UPDATE_HIERARCHY.....	LOIU_DELIVERY_NETWORKF01	2375	64
OIU_VERIFY_PRIMARY_GEO_LOC....	LOIU_DELIVERY_NETWORKF01	744	37
OIU_VERIFY_SECONDARY_GEO_LOC..	LOIU_DELIVERY_NETWORKF01	776	37
POPUP_TO_CONFIRM_STEP.....	LOIU_DELIVERY_NETWORKF01	192	28

SET/GET Params:		Line	Page
OIU_DN.....	LOIU_DELIVERY_NETWORKF01	521	33
	LOIU_DELIVERY_NETWORKF01	1655	52
OIU_DN_GRP.....	LOIU_DELIVERY_NETWORKF01	1664	52
	LOIU_DELIVERY_NETWORKF01	1927	56
OIU_DN_REF.....	LOIU_DELIVERY_NETWORKF01	1683	52
OIU_DN_TYPE_CD.....	LOIU_DELIVERY_NETWORKF01	1675	52
	LOIU_DELIVERY_NETWORKF01	1681	52
OIU_VENTURE_TYPE.....	LOIU_DELIVERY_NETWORKF01	1653	52

DATABASE OPERATIONS

SELECT:		Line	Page
OIUH_PR_DN.....	LOIU_DELIVERY_NETWORKF01	2414	64
OIU_CM_EDIT.....	LOIU_DELIVERY_NETWORKF01	2316	63
OIU_CM_LXPGL_TX.....	LOIU_DELIVERY_NETWORKF01	1077	42
OIU_CM_LXSGL_TX.....	LOIU_DELIVERY_NETWORKF01	1090	42
OIU_DO_JV.....	LOIU_DELIVERY_NETWORKF01	1157	43
	LOIU_DELIVERY_NETWORKF01	1532	50
OIU_PR_DN.....	LOIU_DELIVERY_NETWORKF01	1189	44
	LOIU_DELIVERY_NETWORKF01	1201	44
	LOIU_DELIVERY_NETWORKF01	1657	52
	LOIU_DELIVERY_NETWORKF01	1685	52
	LOIU_DELIVERY_NETWORKF01	2325	63
OIU_PR_DN_GRP.....	LOIU_DELIVERY_NETWORKF01	296	29
	LOIU_DELIVERY_NETWORKF01	1917	56
OIU_PR_DN_GRP_TX.....	LOIU_DELIVERY_NETWORKF01	582	34
	LOIU_DELIVERY_NETWORKF01	1668	52
	LOIU_DELIVERY_NETWORKF01	2255	62
T001.....	LOIU_DELIVERY_NETWORKF01	1149	43
	LOIU_DELIVERY_NETWORKF01	1459	48
T005T.....	LOIU_DELIVERY_NETWORKF01	1067	42

INTERNAL TABLE OPERATIONS

READ TABLE:		Line	Page
GRP_TB.....	LOIU_DELIVERY_NETWORKF01	1923	56
L_SINGLE_DN_SET.....	LOIU_DELIVERY_NETWORKF01	2293	62

```
*———————————————————————————————————————————————————————————————*
*   INCLUDE LOIU_DELIVERY_NETWORKF02                              *
*———————————————————————————————————————————————————————————————*
**********************************************************************
*THE FOLLOWING SUBROUTINES DEAL WITH IDOC OUTPUT CREATION ————————*
*&———————————————————————————————————————————————————————————————*
*&      Form  FILL_CONTROL_SEGMENT
*&———————————————————————————————————————————————————————————————*
*       text - FILL THE TRANSACTIONAL RFC INFORMATION IN THE CONTROL -
*              TABLE
*———————————————————————————————————————————————————————————————*
*      ——>IDOC_CNTRL  text
*      ——>C_IDOC_MESTYP  text
*———————————————————————————————————————————————————————————————*
FORM FILL_CONTROL_SEGMENT TABLES   IDOC_CNTRL STRUCTURE EDIDC
                                   CNTRL_DATA STRUCTURE EDP13.
 LOOP AT CNTRL_DATA.
  CLEAR IDOC_CNTRL.
    IDOC_CNTRL-MANDT    = SY-MANDT.
    IDOC_CNTRL-SERIAL   = SY-DATUM.
    IDOC_CNTRL-SERIAL+8 = SY-UZEIT.
    IDOC_CNTRL-MESTYP   = CNTRL_DATA-MESTYP.
    IDOC_CNTRL-DOCTYP   = C_IDOC_DOCTYP.
    IDOC_CNTRL-DIRECT   = C_IDOC_DIRECT.
    IDOC_CNTRL-RCVPRT   = CNTRL_DATA-RCVPRT.
    IDOC_CNTRL-RCVPRN   = CNTRL_DATA-RCVPRN.
    IDOC_CNTRL-CIMTYP   = CNTRL_DATA-CIMTYP.
    APPEND IDOC_CNTRL.
  ENDLOOP.

ENDFORM.                    " FILL_CONTROL_SEGMENT

*&———————————————————————————————————————————————————————————————*
*&      Form  FILL_DATA_SEGMENT
*&———————————————————————————————————————————————————————————————*
*       text - FILL THE SEGMENT DATA INTO THE EDIDD STRUCTURE
*———————————————————————————————————————————————————————————————*
*      ——>IDOC_DATA  text
*———————————————————————————————————————————————————————————————*
FORM FILL_DATA_SEGMENT USING   I_ROIU_PR_DN STRUCTURE ROIU_PR_DN.
  CLEAR: IDOC_DATA,
         E1OIUDN.
  REFRESH: IDOC_DATA.
  MOVE-CORRESPONDING I_ROIU_PR_DN TO E1OIUDN.
  E1OIUDN-OIU_DATUM     = SY-DATUM.
  E1OIUDN-OIU_UZEIT     = SY-UZEIT.
*For action codes D or U insert the change user into the field 'CRUSER'
*create user.  For action code 'I' leave the create user in this field
   IF NOT E1OIUDN-XXDBACTION = 'I'.
      E1OIUDN-OIU_CRUSER = SY-UNAME.
   ENDIF.

  MOVE E1OIUDN TO IDOC_DATA-SDATA.
  IDOC_DATA-SEGNAM      = 'E1OIUDN'.
  APPEND IDOC_DATA.

ENDFORM.                    " FILL_DATA_SEGMENT

*&———————————————————————————————————————————————————————————————*
*&      Form  SEND_IDOC
*&———————————————————————————————————————————————————————————————*
```

```
*       text SEND THE OUTBOUND IDOC
*———————————————————————————————————————————————————————————————————*
*  ——>  p1        text
*  <——  p2        text
*———————————————————————————————————————————————————————————————————*
FORM SEND_IDOC.
DATA: BEGIN OF COMMUNICATION_IDOC_CONTROL OCCURS 0.
          INCLUDE STRUCTURE EDIDC.
DATA: END OF COMMUNICATION_IDOC_CONTROL .

LOOP AT IDOC_CNTRL.
 CALL FUNCTION 'MASTER_IDOC_DISTRIBUTE'
      EXPORTING
           MASTER_IDOC_CONTROL            = IDOC_CNTRL
*          OBJ_TYPE                       = ''
*          CHNUM                          = ''
      TABLES
           COMMUNICATION_IDOC_CONTROL     = COMMUNICATION_IDOC_CONTROL
           MASTER_IDOC_DATA               = IDOC_DATA
      EXCEPTIONS
           ERROR_IN_IDOC_CONTROL          = 1
           ERROR_WRITING_IDOC_STATUS      = 2
           ERROR_IN_IDOC_DATA             = 3
           SENDING_LOGICAL_SYSTEM_UNKNOWN = 4.

 IF SY-SUBRC <> 0.
  PERFORM SET_RETURN_MESSAGE USING  SY-MSGTY
                                    C_MESSAGEID_CM
                                     SY-MSGNO
                                     SPACE
                                     SPACE
                                     TEXT-IDO
                                     SY-MSGV1
                                     SY-MSGV2
                                     SY-MSGV3
                                     SPACE
                                     SPACE
                                     SPACE.
 ENDIF.

ENDLOOP.
ENDFORM.                    " SEND_IDOC
```

Subroutines:		Line	Page
FILL_CONTROL_SEGMENT.....	LOIU_DELIVERY_NETWORKF02	15	72
FILL_DATA_SEGMENT.....	LOIU_DELIVERY_NETWORKF02	40	72
SEND_IDOC.....	LOIU_DELIVERY_NETWORKF02	67	73
SET_RETURN_MESSAGE.....	LOIU_DELIVERY_NETWORKF02	88	73

External calls

Function modules:		Line	Page
MASTER_IDOC_DISTRIBUTE.....	LOIU_DELIVERY_NETWORKF02	73	73

```
*____________________________________________________________________*
***INCLUDE LOIU_DELIVERY_NETWORKO01 .
*____________________________________________________________________*
*&___________________________________________________________________*
*&      Module  SCREEN_0100_INTIALIZATION  OUTPUT
*&___________________________________________________________________*
*       text
*____________________________________________________________________*
MODULE SCREEN_0100_INTIALIZATION OUTPUT.

  IF W_MODE IS INITIAL.
*   Get the mode from the transaction code
    CALL FUNCTION 'OIU_CM_TXN_CTL'
         EXPORTING
              I_TCODE    = SY-TCODE
         IMPORTING
              E_MCODE    = W_MODE
              E_MCDTXT   = W_TTLTXT
              E_OIUFORMAT = W_FORMAT
        EXCEPTIONS
              NONE_FOUND = 1
              OTHERS     = 2.
    IF SY-SUBRC NE 0.
      PERFORM SET_RETURN_MESSAGE USING SY-MSGTY
                                       SY-MSGID
                                       SY-MSGNO
                                       SPACE
                                       SPACE
                                       SY-MSGV1
                                       SY-MSGV2
                                       SY-MSGV3
                                       SY-MSGV4
                                       SPACE
                                       SPACE
                                       SPACE.
    ENDIF.
  ENDIF.

* Set parameter ids for screen fields
  PERFORM SET_PARAMETER_ID.
* Set the status and the titlebar based on the mode
  REFRESH FCODE_TB.

  FCODE_TB = SY-TCODE.
  APPEND FCODE_TB.

  CALL FUNCTION 'OIUPR_GT_STATUS_CHK'
       TABLES
         t_fcodes = FCODE_TB.

  IF SY-DYNNR = C_SCREEN_1100.
   SET PF-STATUS '0100' EXCLUDING FCODE_TB.
   SET TITLEBAR  '0100' WITH      W_TTLTXT.
  ELSE.
   SET PF-STATUS SY-DYNNR EXCLUDING FCODE_TB.
   SET TITLEBAR  SY-DYNNR WITH      W_TTLTXT.
  ENDIF.

* Turn create by reference field off when not in create mode
  IF W_MODE NE C_CREATE.
    LOOP AT SCREEN.
```

```
      IF SCREEN-GROUP1 = 1.
        SCREEN-ACTIVE = C_OFF_FG.
        MODIFY SCREEN.
      Elseif screen-group2 = 1.
        screen-required = 1.
        modify screen.
      endif.
    ENDLOOP.
  ENDIF.
 IF SY-DYNNR = C_SCREEN_1100.
  PERFORM GET_TYPECD_TEXTS.
 ENDIF.

ENDMODULE.                    " SCREEN_0100_INTIALIZATION  OUTPUT
*&---------------------------------------------------------------------*
*&      Module  SCREEN_0200_INITIALIZATION  OUTPUT
*&---------------------------------------------------------------------*
*       text
*---------------------------------------------------------------------*
MODULE SCREEN_0200_INITIALIZATION OUTPUT.

  REFRESH FCODE_TB.

  IF W_MODE = C_DISPLAY.
    FCODE_TB = C_SAVE.
    APPEND FCODE_TB.
  ENDIF.
  IF W_MODE = C_CREATE.
    FCODE_TB = C_ADMN.
    APPEND FCODE_TB.
  ENDIF.

  FCODE_TB = W_MODE.
  APPEND FCODE_TB.

  CALL FUNCTION 'OIUPR_GT_STATUS_CHK'
       TABLES
         t_fcodes = FCODE_TB.

*  IF SY-DYNNR = C_SCREEN_1200.
*   SET PF-STATUS '0200' EXCLUDING FCODE_TB.
*   SET TITLEBAR  '0200' WITH W_TTLTXT.
*  ELSE.
   SET PF-STATUS SY-DYNNR EXCLUDING FCODE_TB.
   SET TITLEBAR  SY-DYNNR WITH W_TTLTXT.
*  ENDIF.

* Turn all screen fields to display when in display mode
    LOOP AT SCREEN.
      IF SCREEN-GROUP1 = 1 AND W_MODE = C_DISPLAY.
        SCREEN-INPUT = C_OFF_FG.
        MODIFY SCREEN.
      ENDIF.

      IF ROIU_PR_DN-TYPE_CD NE 'P' AND SCREEN-GROUP3 = 3.
        SCREEN-ACTIVE = C_OFF_FG.
        MODIFY SCREEN.
      ENDIF.
    ENDLOOP.


```

```
ENDMODULE.                    " SCREEN_0200_INITIALIZATION  OUTPUT
*&---------------------------------------------------------------------*
*&      Module  SET_TEXTS_SCREEN  OUTPUT
*&---------------------------------------------------------------------*
*       text
*---------------------------------------------------------------------*
MODULE SET_TEXTS_SCREEN OUTPUT.

* Get the country/state/county texts for screen 200
  PERFORM GET_COUNTRY_TEXTS.
   IF SY-DYNNR = C_SCREEN_1100.
*   IF SY-DYNNR = C_SCREEN_1100 OR
*      SY-DYNNR = C_SCREEN_1200.
** Get the type code texts for the delivery network
   PERFORM GET_TYPECD_TEXTS.
  ENDIF.
* Get the company code/JV/gas plant delivery network text
  PERFORM GET_GAS_PLANT_TEXTS.

ENDMODULE.                    " SET_TEXTS_SCREEN  OUTPUT
*&---------------------------------------------------------------------*
*&      Module  POPULATE_COMBOBOXES  OUTPUT
*&---------------------------------------------------------------------*
*  POPULATE PRIMARY AND SECONDARY GEOGRAPHIC LOCATION COMBOBOXES BASED
*  ON WHETHER THE SCREEN FIELD HAS THE SAME VALUE AS WAS USED TO
*  POPULATE THE VALUE LAST.
*---------------------------------------------------------------------*
MODULE POPULATE_COMBOBOXES OUTPUT.

DATA: TABNAME LIKE DD03V-TABNAME.
DATA: FIELDNAME LIKE DD03V-FIELDNAME.
DATA: VALUESET_ID(80) TYPE C.

  IF W_CNTRY <> ROIU_PR_DN-CNTRY_CD.
    W_CNTRY = ROIU_PR_DN-CNTRY_CD.
    W_PRIMARY = ROIU_PR_DN-PRI_GEO_LOC.

    TABNAME = 'ROIU_PR_DN'.
    FIELDNAME = 'PRI_GEO_LOC'.
    CONCATENATE TABNAME '-' FIELDNAME INTO VALUESET_ID.

    CALL FUNCTION 'VRM_DELETE_VALUES'
         EXPORTING
              id           = VALUESET_ID
         EXCEPTIONS
              ID_NOT_FOUND = 1
              OTHERS       = 2
              .

    CALL FUNCTION 'DD_SHLP_COMBOBOX_FILL'
         EXPORTING
              tabname        =  TABNAME
              fieldname      =  FIELDNAME
              valueset_id    =  VALUESET_ID
*        IMPORTING
*             RC             =
*        TABLES
*             VALUE_PAIR_TAB =
              .
    TABNAME = 'ROIU_PR_DN'.
    FIELDNAME = 'SEC_GEO_LOC'.
```

```
184       CONCATENATE TABNAME '-' FIELDNAME INTO VALUESET_ID.
185
186       CALL FUNCTION 'VRM_DELETE_VALUES'
187            EXPORTING
188                 id           = VALUESET_ID
189            EXCEPTIONS
190                 ID_NOT_FOUND = 1
191                 OTHERS       = 2
192                 .
193
194       CALL FUNCTION 'DD_SHLP_COMBOBOX_FILL'
195            EXPORTING
196                 tabname        =  TABNAME
197                 fieldname      =  FIELDNAME
198                 valueset_id    =  VALUESET_ID
199  *         IMPORTING
200  *              RC             =
201  *         TABLES
202  *              VALUE_PAIR_TAB =
203                 .
204     ELSEIF W_PRIMARY <> ROIU_PR_DN-PRI_GEO_LOC.
205
206       W_PRIMARY = ROIU_PR_DN-PRI_GEO_LOC.
207       TABNAME = 'ROIU_PR_DN'.
208       FIELDNAME = 'SEC_GEO_LOC'.
209       CONCATENATE TABNAME '-' FIELDNAME INTO VALUESET_ID.
210
211       CALL FUNCTION 'VRM_DELETE_VALUES'
212            EXPORTING
213                 id           = VALUESET_ID
214            EXCEPTIONS
215                 ID_NOT_FOUND = 1
216                 OTHERS       = 2
217                 .
218
219       CALL FUNCTION 'DD_SHLP_COMBOBOX_FILL'
220            EXPORTING
221                 tabname        =  TABNAME
222                 fieldname      =  FIELDNAME
223                 valueset_id    =  VALUESET_ID
224  *         IMPORTING
225  *              RC             =
226  *         TABLES
227  *              VALUE_PAIR_TAB =
228                 .
229     ENDIF.
230
231  ENDMODULE.                    " POPULATE_COMBOBOXES  OUTPUT
232
```

PBO modules:		Line	Page
POPULATE_COMBOBOXES.....	LOIU_DELIVERY_NETWORKO01	150	78
SCREEN_0100_INTIALIZATION.....	LOIU_DELIVERY_NETWORKO01	9	76
SCREEN_0200_INITIALIZATION....	LOIU_DELIVERY_NETWORKO01	81	77
SET_TEXTS_SCREEN.....	LOIU_DELIVERY_NETWORKO01	129	78

Subroutines:		Line	Page
GET_COUNTRY_TEXTS.....	LOIU_DELIVERY_NETWORKO01	132	78
GET_GAS_PLANT_TEXTS.....	LOIU_DELIVERY_NETWORKO01	140	78
GET_TYPECD_TEXTS.....	LOIU_DELIVERY_NETWORKO01	72	77
	LOIU_DELIVERY_NETWORKO01	137	78
SET_PARAMETER_ID.....	LOIU_DELIVERY_NETWORKO01	40	76
SET_RETURN_MESSAGE.....	LOIU_DELIVERY_NETWORKO01	24	76

Status:		Line	Page
0100.....	LOIU_DELIVERY_NETWORKO01	52	76
?.....	LOIU_DELIVERY_NETWORKO01	55	76
?.....	LOIU_DELIVERY_NETWORKO01	105	77

External calls

Function modules:		Line	Page
DD_SHLP_COMBOBOX_FILL.....	LOIU_DELIVERY_NETWORKO01	172	78
	LOIU_DELIVERY_NETWORKO01	194	79
	LOIU_DELIVERY_NETWORKO01	219	79
OIUPR_GT_STATUS_CHK.....	LOIU_DELIVERY_NETWORKO01	47	76
	LOIU_DELIVERY_NETWORKO01	97	77
OIU_CM_TXN_CTL.....	LOIU_DELIVERY_NETWORKO01	13	76
VRM_DELETE_VALUES.....	LOIU_DELIVERY_NETWORKO01	164	78
	LOIU_DELIVERY_NETWORKO01	186	79
	LOIU_DELIVERY_NETWORKO01	211	79

```
*————————————————————————————————————————————————————————————————————*
***INCLUDE LOIU_DELIVERY_NETWORKI01 .
*————————————————————————————————————————————————————————————————————*
*&————————————————————————————————————————————————————————————————————*
*&      Module  PROCESS_EXIT  INPUT
*&————————————————————————————————————————————————————————————————————*
*       text
*————————————————————————————————————————————————————————————————————*
MODULE PROCESS_EXIT INPUT.

  IF SY-DYNNR = '0100'
   OR SY-DYNNR = C_SCREEN_1100.
    SET SCREEN 0.
    LEAVE SCREEN.
  ELSE.
    IF DELIVERY_NETWORK_TB1 NE ROIU_PR_DN OR W_MODE = C_CREATE.
      CALL FUNCTION 'POPUP_TO_CONFIRM_LOSS_OF_DATA'
           EXPORTING
                TEXTLINE1 = TEXT-010
                TEXTLINE2 = TEXT-017
                TITEL     = TEXT-009
           IMPORTING
                ANSWER    = W_ANSWER.
      IF W_ANSWER = C_YES.
*—— begin change note 196947 05/21/2001 lk
*—— the table should not be locked for all the networks
*—— when creating a external delivery network.
      PERFORM unlock_delivery_number.
*—— end change note 196947 05/21/2001 lk
       PERFORM CLEAR_INFORMATION.
*       IF SY-DYNNR = C_SCREEN_1200.
*         SET SCREEN 1100.
*         LEAVE SCREEN.
*        ELSE.
         SET SCREEN 100.
         LEAVE SCREEN.
*        ENDIF.
      ENDIF.
    ELSE.
      PERFORM CLEAR_INFORMATION.
*      IF SY-DYNNR = C_SCREEN_1200.
*       SET SCREEN 1100.
*       LEAVE SCREEN.
*      ELSE.
       SET SCREEN 100.
       LEAVE SCREEN.
*      ENDIF.
    ENDIF.
  ENDIF.

ENDMODULE.                    " PROCESS_EXIT  INPUT

*&————————————————————————————————————————————————————————————————————*
*&      Module  USER_COMMAND_0100  INPUT
*&————————————————————————————————————————————————————————————————————*
*       text
*————————————————————————————————————————————————————————————————————*
MODULE USER_COMMAND_0100 INPUT.

  CASE W_FCODE.
    WHEN C_ENTER.
```

```
      IF W_MODE = C_UPDATE.
*       Set the DB action for updates and lock the number from changes
        ROIU_PR_DN-XXDBACTION = 'U'.
        PERFORM LOCK_DELIVERY_NUMBER.
*       Move the original dn info into 2 temp internal tables for later
*       comparisons to see if changes have occured
        DELIVERY_NETWORK_TB1 = ROIU_PR_DN.
*       delivery_network_tb2 = delivery_network_tb1.
        APPEND DELIVERY_NETWORK_TB1.
*       APPEND delivery_network_tb2.
      ENDIF.
      IF SY-DYNNR = C_SCREEN_1100.
**     SET SCREEN 1200.
       SET SCREEN 200.
       LEAVE SCREEN.
      ELSE.
       SET SCREEN 200.
       LEAVE SCREEN.
      ENDIF.
  ENDCASE.

ENDMODULE.                    " USER_COMMAND_0100  INPUT
*&---------------------------------------------------------------------*
*&      Module  USER_COMMAND_0200  INPUT
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
MODULE USER_COMMAND_0200 INPUT.

  CASE W_FCODE.
    WHEN C_CREATE OR C_UPDATE OR C_DISPLAY.
*     Leave to the beginning of a new type of transaction
      IF ( DELIVERY_NETWORK_TB1 NE ROIU_PR_DN  AND
           W_MODE = C_UPDATE )                 OR
           W_MODE = C_CREATE.
        CALL FUNCTION 'POPUP_TO_CONFIRM_LOSS_OF_DATA'
             EXPORTING
                  TEXTLINE1 = TEXT-010
                  TEXTLINE2 = TEXT-017
                  TITEL     = TEXT-009
             IMPORTING
                  ANSWER    = W_ANSWER.
      ELSE.
        W_ANSWER = C_YES.
      ENDIF.
      IF W_ANSWER = C_YES.
        PERFORM CLEAR_INFORMATION.
        W_MODE = W_FCODE.
        PERFORM LEAVE_TO_TRANSACTION USING W_MODE.
      ENDIF.
    WHEN C_OTHR.
*     Leave to the beginning of the same type of transaction
      PERFORM LEAVE_TO_TRANSACTION USING W_MODE.
    WHEN C_ADMN.
*     Call the administrative data popup window
      PERFORM ADMINISTRATIVE_DATA.
    WHEN C_OBJC.
*     Check for the change document
      OBJECTID = ROIU_PR_DN-DN_NO.
      SUBMIT RSSCD100 WITH OBJEKT   EQ 'OIU_DN'
                      WITH OBJEKTID EQ OBJECTID
```

```
                        WITH AENDERER EQ SPACE
                        WITH DATUM    EQ SPACE
                        WITH ZEIT     EQ SPACE
                        WITH NUMMER   EQ SPACE
                        WITH TABNAME  EQ SPACE
                        WITH TABKEY   EQ SPACE
                        WITH KEY_EXP  EQ 'X'
                        AND  RETURN.
     WHEN C_LONG.
*      Calls the long text processing function
       PERFORM PREPARE_LONG_TEXT.
     WHEN C_SAVE.
**       perform verify_entity_name.
         PERFORM CHECK_FOR_GAS_PLANT.
         IF NOT ROIU_PR_DN-GPLT_DN_NO IS INITIAL.
            PERFORM CHECK_VALID_GASPLANT_NTWK_ID.
         ENDIF.
       check W_FCODE = c_save.
*      Stop processing when save is selected but no changes have occured
       IF DELIVERY_NETWORK_TB1 EQ ROIU_PR_DN
          AND W_LONG_TEXT_WRITTEN_FG IS INITIAL.
*        Send user a message when no changes have occured on save
         PERFORM NO_CHANGES_MADE.
       ENDIF.
       CASE W_MODE.
         WHEN C_CREATE.
           PERFORM CREATE_DELIVERY_NETWORK.
         WHEN C_UPDATE.
           PERFORM UPDATE_DELIVERY_NETWORK.
       ENDCASE.
     WHEN C_BACK OR C_EXIT.
*      Checks to see if there are any changes if so popup to save is
*      activated.
       PERFORM CHECK_FOR_CHANGES.
* Change to delivery network dated transaction
     WHEN C_FCODE_DND1 OR C_FCODE_DND2 OR C_FCODE_DND3 OR C_FCODE_DND6.
       PERFORM CHANGE_TO_DN_DATED.
   ENDCASE.

   IF W_FCODE = C_BACK AND W_ANSWER <> C_CANCEL.
     IF SY-CALLD = C_TRUE  OR  SY-BINPT = C_TRUE.
       LEAVE TO SCREEN 0.
     ELSE.
       LEAVE TO TRANSACTION SY-TCODE.
     ENDIF.

   ELSEIF W_FCODE = C_EXIT AND W_ANSWER <> C_CANCEL.
       PERFORM CLEAR_INFORMATION.
       SET SCREEN 0.
       LEAVE SCREEN.
   ENDIF.

ENDMODULE.                    " USER_COMMAND_0200  INPUT
*&---------------------------------------------------------------------*
*&      Module  CHECK_NUMBER_RANGE  INPUT
*&---------------------------------------------------------------------*
*       text
*----------------------------------------------------------------------*
MODULE CHECK_NUMBER_RANGE INPUT.

   IF W_MODE = C_CREATE.
```

```
*   Check to see if the number range group exists
*    PERFORM CHECK_FOR_NUMBER.
*   Get the number range information in order to get the number range
    PERFORM GET_NUMBER_RANGE_INFO.
*   If the number range is external lock the delivery network number
    IF W_INTERVAL_DS-EXTERNIND = C_FLAG_ON.
      PERFORM LOCK_DELIVERY_NUMBER.
    ENDIF.
*   Checks to see if the proper fields are populated based on internal
*   and external number ranges
    PERFORM CHECK_FOR_VALIDITY.
*   If the indicator is not external set del network number = INTER-001
    IF W_INTERVAL_DS-EXTERNIND NE C_TRUE.
      ROIU_PR_DN-DN_NO = C_INTERNAL.
    ENDIF.
  ENDIF.

ENDMODULE.                    " CHECK_NUMBER_RANGE  INPUT
*&---------------------------------------------------------------------*
*&      Module  CHECK_DELIVERY_NETWORK_DATA  INPUT
*&---------------------------------------------------------------------*
*       text
*---------------------------------------------------------------------*
MODULE CHECK_DELIVERY_NETWORK_DATA INPUT.

  CASE W_MODE.
    WHEN C_CREATE.
*     Check for a duplicate delivery network for external number assign
      PERFORM CHECK_FOR_DUPLICATE.
*     Check for a duplicate DN in Razorfish OIUH_PR_DN table    "N 17322
81
      PERFORM CHECK_RAZORFISH_DN_DUPLICATE.                     "N 17322
81
*     When using create by reference get the reference del network
      IF NOT ROIU_PR_DN-REF_DN_NO IS INITIAL.
        W_REF_FG = C_FLAG_ON.
*       get delivery network detail info for create by reference
        PERFORM GET_REF_DELIVERY_NETWORK USING ROIU_PR_DN-REF_DN_NO
                                               TEXT-001
                                               TEXT-002
                                               ROIU_PR_DN-REF_DN_NO
                                               W_REF_FG.
      ELSE.
        PERFORM DEFAULT_HIER_NODE.
      ENDIF.
    WHEN OTHERS.
      CLEAR W_REF_FG.
*     Get the delivery network detail information
      PERFORM GET_REF_DELIVERY_NETWORK USING ROIU_PR_DN-DN_NO
                                             TEXT-002
                                             ROIU_PR_DN-DN_NO
                                             SPACE
                                             W_REF_FG.
  ENDCASE.
ENDMODULE.                    " CHECK_DELIVERY_NETWORK_DATA  INPUT
*&---------------------------------------------------------------------*
*&      Module  CHECK_GAS_PLANT_NETWORK  INPUT
*&---------------------------------------------------------------------*
*       text
*---------------------------------------------------------------------*
*MODULE CHECK_GAS_PLANT_NETWORK INPUT.
```

```
*
*  IF W_MODE NE C_DISPLAY.
**   Validates required fields (company code/Joint Venture) when
**   gas plant delivery network type is selected
*    PERFORM CHECK_FOR_GAS_PLANT.
**   When the gas plant delivery network is filled make sure it is valid
**   gas plant network
*    IF NOT ROIU_PR_DN-GPLT_DN_NO IS INITIAL.
*      PERFORM CHECK_VALID_GASPLANT_NTWK_ID.
*    ENDIF.
*  ENDIF.
*
*ENDMODULE.                    " CHECK_GAS_PLANT_NETWORK  INPUT
*&---------------------------------------------------------------------*
*&      Module  CHECK_CNTRY_REGION_CITY  INPUT
*&---------------------------------------------------------------------*
*       text
*---------------------------------------------------------------------*
MODULE CHECK_CNTRY_REGION_CITY INPUT.

  IF W_MODE NE C_DISPLAY.
*   Check if country has a required city/state etc.
    CHECK W_FCODE <> C_FCODE_LOC.
    PERFORM CHECK_COUNTRY_AND_REGION.
  ENDIF.

ENDMODULE.                    " CHECK_CNTRY_REGION_CITY  INPUT
*&---------------------------------------------------------------------*
*&      Module  SET_FUNCTION_CODE  INPUT
*&---------------------------------------------------------------------*
*       text
*---------------------------------------------------------------------*
MODULE SET_FUNCTION_CODE INPUT.

  CLEAR W_FCODE.
  W_FCODE = OK_CODE.

ENDMODULE.                    " SET_FUNCTION_CODE  INPUT
*&---------------------------------------------------------------------*
*&      Module  CHECK_COMBOBOX_STATUS  INPUT
*&---------------------------------------------------------------------*
*  CALL PERFORM TO RESET COMBOBOXES BASED ON VALUE CHANGES.
*---------------------------------------------------------------------*
module check_combobox_status input.
* IF SY-DYNNR <> C_SCREEN_1200.
  PERFORM CHECK_COMBOBOX_STATUS.
* ENDIF.
endmodule.                    " CHECK_COMBOBOX_STATUS  INPUT


*&---------------------------------------------------------------------*
*&      Module  CHECK_SETID  INPUT
*&---------------------------------------------------------------------*
* VALIDATE THAT THE SETID THAT IS INPUT EXISTS IN THE STANDARD HIERARCHY
* FOR THE DELIVERY NETWORK SET CLASS.
*---------------------------------------------------------------------*
module check_setid input.
* IF SY-DYNNR = '1200'.
*  PERFORM CHECK_DN_SETID USING ROIU_PR_DN-DN_SET.
* ENDIF.
  PERFORM CHECK_SETID USING ROIU_PR_DN-DN_SET.
```

```
304
305 endmodule.                  " CHECK_SETID  INPUT
306
307 *&———————————————————————————————————————————————————————————————*
308 *&      Module  HELP_DN_SET  INPUT
309 *&———————————————————————————————————————————————————————————————*
310 *       text
311 *———————————————————————————————————————————————————————————————*
312 module help_dn_set input.
313
314
315   W_PROGRAM = SY-CPROG.
316   W_DYNNR   = SY-DYNNR.
317
318   IF W_MODE <> C_CREATE AND
319      W_MODE <> C_UPDATE.
320     W_DISPLAY = C_FLAG_ON.
321   ELSE.
322     CLEAR W_DISPLAY.
323   ENDIF.
324
325   CALL FUNCTION 'OIU_HELP_STD_HIER_NODES'
326        EXPORTING
327             fieldname    = C_DN_SET_FNAME
328             program      = W_PROGRAM
329             dynnr        = W_DYNNR
330             setclass     = C_SETCLASS
331             DISPLAY_ONLY = W_DISPLAY
332             .
333
334 endmodule.                  " HELP_MP_SET  INPUT
335
336 *&———————————————————————————————————————————————————————————————*
337 *&      Module  CHECK_DNDE  INPUT
338 *&———————————————————————————————————————————————————————————————*
339 *       text
340 *———————————————————————————————————————————————————————————————*
341 MODULE CHECK_DNDE INPUT.
342 * IF SY-DYNNR = '1200'.
343 ** Check that required field description is populated
344 *  PERFORM CHECK_DESCRIPTION.
345 * ENDIF.
346 ENDMODULE.                  " CHECK_DNDE  INPUT
347
348 *&———————————————————————————————————————————————————————————————*
349 *&      Module  CHECK_TOL  INPUT
350 *&———————————————————————————————————————————————————————————————*
351 *       text
352 *———————————————————————————————————————————————————————————————*
353 MODULE CHECK_TOL INPUT.
354   perform check_allocation_tolerance.
355 ENDMODULE.                  " CHECK_TOL  INPUT
356
```

PAI modules:		Line	Page
CHECK_CNTRY_REGION_CITY.....	LOIU_DELIVERY_NETWORKI01	261	86
CHECK_COMBOBOX_STATUS.....	LOIU_DELIVERY_NETWORKI01	286	86
CHECK_DELIVERY_NETWORK_DATA...	LOIU_DELIVERY_NETWORKI01	207	85
CHECK_DNDE.....	LOIU_DELIVERY_NETWORKI01	341	87
CHECK_NUMBER_RANGE.....	LOIU_DELIVERY_NETWORKI01	181	84
CHECK_SETID.....	LOIU_DELIVERY_NETWORKI01	299	86
CHECK_TOL.....	LOIU_DELIVERY_NETWORKI01	353	87
HELP_DN_SET.....	LOIU_DELIVERY_NETWORKI01	312	87
PROCESS_EXIT.....	LOIU_DELIVERY_NETWORKI01	9	82
SET_FUNCTION_CODE.....	LOIU_DELIVERY_NETWORKI01	275	86
USER_COMMAND_0100.....	LOIU_DELIVERY_NETWORKI01	58	82
USER_COMMAND_0200.....	LOIU_DELIVERY_NETWORKI01	89	83

Subroutines:		Line	Page
ADMINISTRATIVE_DATA.....	LOIU_DELIVERY_NETWORKI01	117	83
CHANGE_TO_DN_DATED.....	LOIU_DELIVERY_NETWORKI01	159	84
CHECK_ALLOCATION_TOLERANCE....	LOIU_DELIVERY_NETWORKI01	354	87
CHECK_COMBOBOX_STATUS.....	LOIU_DELIVERY_NETWORKI01	288	86
CHECK_COUNTRY_AND_REGION.....	LOIU_DELIVERY_NETWORKI01	266	86
CHECK_FOR_CHANGES.....	LOIU_DELIVERY_NETWORKI01	156	84
CHECK_FOR_DUPLICATE.....	LOIU_DELIVERY_NETWORKI01	212	85
CHECK_FOR_GAS_PLANT.....	LOIU_DELIVERY_NETWORKI01	136	84
CHECK_FOR_VALIDITY.....	LOIU_DELIVERY_NETWORKI01	194	85
CHECK_RAZORFISH_DN_DUPLICATE..	LOIU_DELIVERY_NETWORKI01	214	85
CHECK_SETID.....	LOIU_DELIVERY_NETWORKI01	303	86
CHECK_VALID_GASPLANT_NTWK_ID..	LOIU_DELIVERY_NETWORKI01	138	84
CLEAR_INFORMATION.....	LOIU_DELIVERY_NETWORKI01	30	82
	LOIU_DELIVERY_NETWORKI01	40	82
	LOIU_DELIVERY_NETWORKI01	108	83
	LOIU_DELIVERY_NETWORKI01	170	84
CREATE_DELIVERY_NETWORK.....	LOIU_DELIVERY_NETWORKI01	149	84
DEFAULT_HIER_NODE.....	LOIU_DELIVERY_NETWORKI01	225	85
GET_NUMBER_RANGE_INFO.....	LOIU_DELIVERY_NETWORKI01	187	85
GET_REF_DELIVERY_NETWORK.....	LOIU_DELIVERY_NETWORKI01	219	85
	LOIU_DELIVERY_NETWORKI01	230	85
LEAVE_TO_TRANSACTION.....	LOIU_DELIVERY_NETWORKI01	110	83
	LOIU_DELIVERY_NETWORKI01	114	83
LOCK_DELIVERY_NUMBER.....	LOIU_DELIVERY_NETWORKI01	65	83
	LOIU_DELIVERY_NETWORKI01	190	85
NO_CHANGES_MADE.....	LOIU_DELIVERY_NETWORKI01	145	84
PREPARE_LONG_TEXT.....	LOIU_DELIVERY_NETWORKI01	133	84
UNLOCK_DELIVERY_NUMBER.....	LOIU_DELIVERY_NETWORKI01	28	82
UPDATE_DELIVERY_NETWORK.....	LOIU_DELIVERY_NETWORKI01	151	84

Screens:		Line	Page
100.....	LOIU_DELIVERY_NETWORKI01	35	82
	LOIU_DELIVERY_NETWORKI01	45	82
200.....	LOIU_DELIVERY_NETWORKI01	75	83
	LOIU_DELIVERY_NETWORKI01	78	83

External calls

Function modules:		Line	Page
OIU_HELP_STD_HIER_NODES.....	LOIU_DELIVERY_NETWORKI01	325	87
POPUP_TO_CONFIRM_LOSS_OF_DATA.	LOIU_DELIVERY_NETWORKI01	17	82
	LOIU_DELIVERY_NETWORKI01	97	83

Reports:		Line	Page
RSSCD100.....	LOIU_DELIVERY_NETWORKI01	121	83